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High-low-frequency feature-guided diffusion model for hyperspectral anomaly detection

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ABSTRACT

Recently, diffusion models have been introduced into the field of hyperspectral anomaly detection (HAD) and have achieved excellent results. However, during the forward diffusion, noise injection degrades the physical structure of the input hyperspectral image (HSI), leading to degraded reconstruction accuracy. In addition, anomaly targets may be reconstructed due to the strong generative capability of diffusion models. The small difference between anomaly targets and background in the reconstruction error constrains the anomaly detection performance. To address these problems, we proposed a high-low-frequency feature-guided diffusion model for HAD. First, in the forward diffusion process, a low-frequency feature-guided module (LFGM) was proposed to utilize the structural features of the HSI through dynamic threshold segmentation wavelet transform (WT). Then, in the reverse diffusion process, a high-frequency and anomaly feature-guided module (HAFGM) was proposed by combining high-frequency filter and spectral correlation calculation to guide the network to generate accurate HSI and anomaly targets. Finally, the generated HSI was fused with the original input HSI to extract anomaly detection result. Experimental results demonstrate that compared with other state-of-the-art methods, the proposed method obtains satisfactory detection results on five real-world datasets.

1. Introduction

Hyperspectral remote sensing can provide hyperspectral images (HSIs) containing rich spectral and spatial information by combining two-dimensional spatial imaging and spectral technology [1–2]. Due to the high spectral resolution and spatial-spectral integration of the HSI, it has demonstrated significant advantages in target detection and recognition [3]. Among them, hyperspectral anomaly detection (HAD) aims to identify pixels whose spectra differ significantly from the background without requiring prior spectral information [4]. At present, HAD has a wide range of applications in environmental monitoring, agricultural assessment, and mineral exploration [5–6].

Over the past thirty years, a large number of HAD methods have been proposed [7]. The existing HAD methods are divided into two categories: traditional and deep learning [8]. Traditional methods are

divided into two categories: statistics-based and representation-based methods [9]. The statistics-based methods model the background distribution of HSIs, considering pixels that deviate from this distribution as anomaly targets [10]. However, the actual scene complexity often leads to inaccurate background modeling, causing detection bias as the assumed distribution deviates from actual background characteristics. The representation-based methods represent each pixel of the HSI through a constructed overcomplete dictionary [11]. Anomaly targets are extracted from reconstruction errors. Although representation-based methods demonstrate excellent performance, the detection accuracy depends on the quality of the dictionary and the precision of representation. Moreover, the iterative convergence in computation has randomness, and the experimental results may be inconsistent [12–13].

Due to the lack of prior spectral information, deep learning-based HAD methods extract deep features from HSIs through unsupervised

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networks. Anomaly probabilities are then estimated based on reconstruction errors [14]. According to the model structure, existing deep learning-based methods can be divided into autoencoder (AE)-based, generative adversarial network (GAN)-based, and diffusion model-based methods [15]. The AE-based methods reconstruct HSIs through an encoder-decoder architecture. The reconstruction error between input HSI and reconstructed HSI is used to extract the anomaly targets [16]. Currently, various AE variants have been proposed to improve anomaly detection performance. However, AE-based methods struggle to accurately reconstruct complex background and can be biased by anomaly targets during training. The GAN-based methods leverage the adversarial training between generators and discriminators to model hyperspectral data distribution, enabling anomaly detection through residual analysis [17]. Although GAN has shown excellent potential in HAD, there are some limitations. Due to the absence of prior spectral knowledge, the GAN-based methods struggle to acquire uncontaminated training data, as anomaly targets and background coexist in the training data. Additionally, the training of the GAN is unstable.

As emerging generative models, diffusion models have demonstrated remarkable performance in image generation and restoration by progressively learning data distribution through iterative denoising [18]. Currently, diffusion models have been successfully applied to HAD, demonstrating competitive detection performance [19–20]. The diffusion model-based HAD methods characterize background distributions by simulating the forward and reverse diffusion processes of HSIs, enabling identification of anomaly targets that deviate from these distributions. Compared to other approaches, the diffusion model-based HAD methods can accurately capture complex HSI data distributions through iterative noise suppression, while maintaining HSI generation capability.

In the HAD methods, diffusion models can generate HSIs and use the reconstruction error as anomaly score. However, during the forward diffusion process, progressive noise addition may degrade the physical structure of the input HSI. Consequently, this structural degradation can prevent diffusion models from completely learning HSI structural features during training, leading to information loss. Therefore, the generated HSIs may not be accurate. Furthermore, due to anomaly targets only occupy a small portion of the pixels in the HSI, the diffusion model-based HAD methods conventionally utilize entire HSIs as training data. These methods assume that the diffusion models learn background features rather than anomaly features. Consequently, the diffusion models can accurately generate background but fails to reproduce anomaly targets, enabling anomaly detection through the reconstruction error. Nevertheless, owing to the strong generative capability, diffusion models may still generate some anomaly targets, despite the number of pixels of anomaly targets is small. This limits the performance of HAD methods.

To solve the above problems, this study proposed a HAD method based on high-low-frequency feature-guided diffusion model. First, by combining the dynamic threshold segmentation wavelet transform (WT), we embedded a low-frequency feature-guided module (LFGM) in the forward diffusion process. This ensures that the network can learn noise patterns while introducing global structural information of the input HSI. Then, during the reverse diffusion process, we proposed a high-frequency and anomaly feature-guided module (HAFGM) that integrates high-frequency filter and spectral correlation calculation. By embedding extracted high-frequency detail and anomaly features from the input HSI through the HAFGM, the network is guided to generate both the HSI and anomaly targets. Finally, anomaly targets are extracted by computing the dot product of the spectral sum between the input HSI and the generated HSI. The main contributions of this study are as follows.

- (1) We proposed a HAD framework based on high-low-frequency feature-guided diffusion model. An LFGM was designed during the forward diffusion process. This module improves the ability of

the network to model the global image structure while learning noise patterns by incorporating the global structural information from the input HSI.

- (2) In the reverse diffusion process, the high-frequency detailed and anomaly target features of the HSI extracted through the constructed HAFGM are used as prior features to guide the network to accurately generate the HSI and anomaly targets, respectively. Unlike the most of the methods that rely on reconstruction errors, we detected anomaly targets by fusing input and generated HSIs. This strategy overcomes the limitation caused by subtle spectral differences between anomaly targets and background in the reconstruction errors.
- (3) We demonstrated the superiority of the proposed approach via extensive experiments on five hyperspectral datasets, validating improvements across multiple AUC-based metrics and qualitative visualizations. Experimental results show that compared with other state-of-the-art methods, the proposed method achieves satisfactory detection results in both qualitative and quantitative aspects.

The other sections of this study are arranged as follows. Section 2 introduces the related work. Section 3 provides a detailed description of the proposed method. Section 4 compares and analyzes the experimental results. Section 5 provides a summary of this study.

2. Related work

2.1. Traditional HAD method

The most classic method based on statistics is the anomaly detection method proposed by Reed-Xiaoli (RX) [21]. The RX method, also called global RX (GRX), assumes that the background follows a multivariate Gaussian distribution, classifying pixels that deviate from it as anomaly targets. Additionally, different GRX-based variants have been developed. For example, local RX (LRX) [22], weighted RX (WRX) [23], subspace RX (SSRX) [24], kernel RX (KRX) [25], two-stage generalized likelihood ratio test (2S-GLRT) [26], and effective anomaly space methods [27–28]. The statistics-based methods describe the background of the HSI through a hypothetical background model, which can quickly detect anomaly targets.

The representation-based HAD method includes sparse representation, collaborative representation, low-rank and sparse matrix decomposition, low-rank representation, and tensor decomposition. The sparse representation-based HAD method assumes that the tested pixels can be sparsely represented by a few atoms from an overcomplete dictionary [29]. Chen et al. pioneered the application of sparse representation theory to hyperspectral target detection [30–31]. Different from sparse representation, the collaborative representation-based HAD method proposed by Li et al. employs all atoms in an overcomplete dictionary to reconstruct the tested pixel [32–34]. The low-rank and sparse matrix decomposition (LRaSMD) method [35] separates the HSI into low-rank component and sparse component, leveraging the low-rank attributes of the background and the sparse attributes of anomaly targets. To improve detection performance, different improvement methods have been proposed. For example, the low-rank and sparse matrix decomposition based Mahalanobis distance method proposed by Zhang et al. [36] and the low-rank and sparse decomposition with mixture of gaussian proposed by Li et al. [37]. Due to the LRaSMD method fails to account for background diversity, low-rank representation models are introduced into HAD. Typical methods include low-rank and sparse representation (LRASR) [38], merging total variation into low rank representation (MTVLRR) [39], learnable background endmember with subspace representation (LEBSR) [40], and principal component analysis and tensor low-rank and sparse representation (PCA-TLRSR) [41]. Tensor decomposition-based HAD models hyperspectral data as three-order tensors, exploiting tensor decomposition to separate low-rank

background components. Li et al. [42] proposed a prior-based tensor approximation method [43] to consider the tensor structure information of HSIs.

2.2. Ae-based HAD method

AE have been applied to HAD because of the strong unsupervised feature extraction capabilities. Early approaches used AE reconstruction errors to identify anomaly targets [44–46]. However, studies revealed that AE could reconstruct both background and anomaly targets spectra, limiting the detection performance. Therefore, numerous enhanced methods have been proposed to address these limitations. To address the lack of prior information, Xie et al. proposed semisupervised background estimation model based on AE and adversarial-learning [47]. This method only uses background spectral samples during training. Fan et al. proposed robust graph AE (RGAE) framework incorporating graph regularization terms and $L_{2,1}$ norm constraint to effectively capture the spatial structural information in HSIs [48]. Wang et al. proposed autonomous HAD (Auto-AD) method based on full convolution networks, removing dependency on manual parameter tuning [49]. Xiang et al. proposed guided AE detector (GAED) by embedding a guided module in AE to suppress the reconstruction of anomaly targets [50]. Ning et al. developed optimal transportation-based cluster memory augmented deep AE to suppress the reconstruction of anomaly targets while enhancing the representation of background feature [51]. To consider the background diversity, Cheng et al. proposed a deep feature aggregation network (DFAN) by adding an adaptive aggregation model [52]. Wang et al. proposed a blind block reconstruction network (BockNet) incorporating a guard window mechanism to reduce anomaly contributions during background reconstruction [53]. Meanwhile, Wang et al. proposed a nonlocal and local feature-coupled self-supervised network (NL2Net) to comprehensively exploit diverse HSI features [54]. To extract nonlinear and spatial features from HSIs, Wu et al. proposed the transformer-based AE framework integrating transformer architecture with AE networks [55]. Liu et al. proposed a self-supervised multi-scale network (MSNet) that employs a separation training strategy via self-supervised learning to suppress the reconstruction of anomaly targets [56].

2.3. Gan-based HAD method

GAN, as another generative model, has demonstrated outstanding performance in HAD. To extract the background features of HSIs and improve anomaly detection performance, Wang et al. proposed a HAD method based on variant background inference and generative adversarial [57]. Shanmugam et al. proposed a dual-discriminator conditional GAN network for HAD, employing the hybrid manta ray serving optimization and volcano entropy methods to enhance background-anomaly reconstruction separation [58]. To reduce the contamination of background features by anomaly targets, Jiang et al. proposed a semi-supervised spectral learning method based on GAN through combining filter and background suppression techniques [59]. To improve the detection performance in open scenarios cross domain detection, Wang et al. proposed continual learning-based capsule differential GAN [60]. This method improves the practical application of HAD in cross domain scenarios.

In addition to AE-based and GAN-based HAD methods, there are also some other deep learning-based methods. For example, anomaly detector based on convolutional neural network [61], denoising convolutional neural network regularization-based anomaly detector [62], transferable network with Siamese architecture-based HAD [63], and adaptive graph frequency location-based HAD [64].

2.4. Diffusion model-based HAD method

Recently, diffusion models have demonstrated powerful generative

capabilities in modeling complex HSI distributions, providing novel solutions for HAD. To enhance anomaly-background separation and improve HAD generalization, Ma et al. proposed a background suppression diffusion model (BSDM) by introducing denoising network [19]. Experimental results demonstrated that the BSDM methods achieved superior detection performance. To mitigate anomaly interference in HSIs, Chen et al. proposed a HAD method based on dual-window spectral diffusion (DWSDiff), integrating three key components: dual window strategy, spectral diffusion process, and reverse reconstruction process [20]. Wu et al. proposed a HAD method based on diffusing background dictionary by combining diffusion model and LRR model [65]. This method constructs a background dictionary using a diffusion model, effectively minimizing the contamination from both anomaly and noise for the background. To exploit spatial features of HSIs, Yang et al. proposed an improved background suppression diffusion model for HAD method. This improved approach incorporates weighted output allocation and nonlinear mapping [66].

The indispensability of diffusion models lies in their support for multi-source conditional guidance, the ability to generate anomaly perception, ease of control in the generation process, and the foundation of our proposed detection approach. First, the proposed method injects both low-frequency guided feature and sampling guided features to jointly guide the generation of HSI structures and anomaly targets. Second, the iterative mechanism of gradual denoising allows for fine-grained control over the generated content. This allows the model to synthesize anomaly targets in designated areas while maintaining the global structure. Third, unlike the AE or GAN methods that typically suppress anomaly target reconstruction, the diffusion model in the proposed method is explicitly guided to generate anomaly targets. Finally, the ability to generate a complex and condition guided HSI with controllable differences from the input HSI is the foundation of the proposed fusion detection strategy.

3. Proposed methodology

The overall framework of the proposed HAD method based on high-low-frequency feature-guided diffusion model is shown in Fig. 1. In this section, we will provide a detailed introduction of the proposed method from four aspects: diffusion model, LFGM, HAFGM, and model training.

3.1. Diffusion model

As probabilistic diffusion process-based generative models, Diffusion models have been successfully applied to diverse image processing tasks. At present, diffusion models have gradually been adopted for the HAD. However, the forward diffusion process progressively degrades the structural integrity of the input HSI through noise perturbation, potentially affecting reconstruction accuracy. In addition, the diffusion models may generate anomaly targets during the reverse diffusion process, reducing the difference in the reconstruction error between anomaly targets and background. Hence, the detection probability is reduced.

To address the aforementioned problems, we proposed a diffusion model guided by high-low-frequency features, as shown in Fig. 1. This model extracts low-frequency features from the input HSI and various noisy HSIs during the forward diffusion process through the LFGM. The forward diffusion process is guided to learn image global structural information and noise patterns through the extracted low-frequency features. During the reverse diffusion process, the HAFGM is used to extract high-frequency detailed and anomaly features from the input HSI. Among them, the extracted high-frequency detailed feature guides the network to generate accurate HSI, while anomaly feature guides the network to generate anomaly targets. Finally, unlike the method of extracting anomaly targets by reconstruction error, we added the input HSI and the generated HSI to extract anomaly detection result. This approach avoids the limited distinguishability of the reconstruction

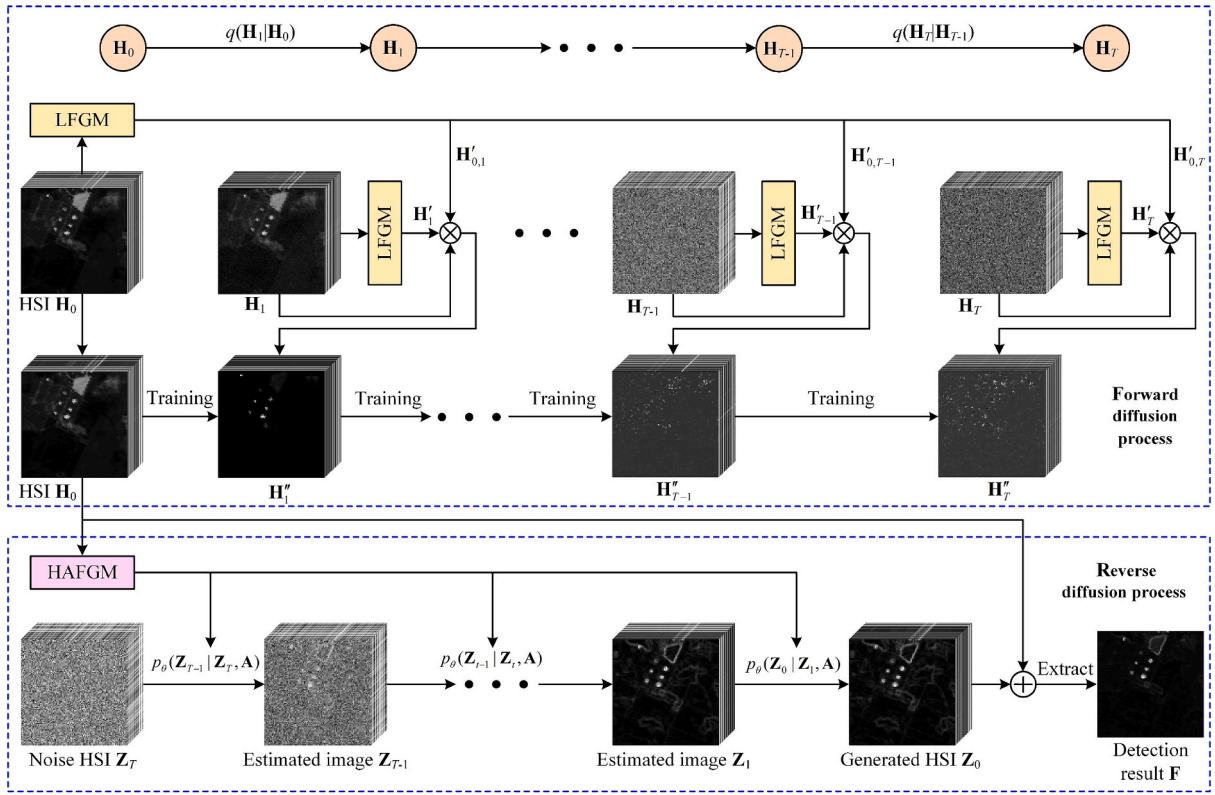


Fig. 1. Overall architecture of the proposed high-low-frequency feature-guided diffusion model.

errors caused by the ability of the diffusion models to partially regenerate anomaly targets.

We assume that the input HSI is represented as $\mathbf{H}_0 \in \mathbb{R}^{M \times N \times L}$, where $M \times N$ is the number of pixels in the HSI, and L is the number of bands in the HSI. The forward diffusion process sequentially adds Gaussian noise to the input HSI to obtain noisy HSIs $\mathbf{H}_1, \mathbf{H}_2, \dots, \mathbf{H}_t, \dots, \mathbf{H}_T$, where T represents time step. The forward diffusion process is represented as

$$q(\mathbf{H}_{1:T}|\mathbf{H}_0) = \prod_{t=1}^T q(\mathbf{H}_t|\mathbf{H}_{t-1}) \quad (1)$$

$$q(\mathbf{H}_t|\mathbf{H}_{t-1}) = \mathcal{N}(\mathbf{H}_t; \sqrt{1 - \beta_t}\mathbf{H}_{t-1}, \beta_t\mathbf{I}) \quad (2)$$

$$\alpha_t = 1 - \beta_t \quad (3)$$

where $q(\cdot)$ is the forward diffusion process, $\mathbf{H}_t$ and $\mathbf{H}_{t-1}$ represent the HSIs with added noise at the t -th and $t-1$ -th time step, respectively, β_t represents the intensity parameter of the added noise and satisfies $0 < \beta_1 < \beta_2 < \dots < \beta_t < \dots < \beta_T < 1$, α_t represents the retention ratio of the original input HSI after adding noise at the t -th time step, and $\mathbf{I}$ represents the identity matrix. To make the above equations computable, through the reparameterization theorem, the process of adding noise at the t -th time step is expressed as

$$\mathbf{H}_t = \sqrt{\alpha_t}\mathbf{H}_{t-1} + \sqrt{1 - \alpha_t}\epsilon_\theta, \epsilon_\theta \sim \mathcal{N}(0, \mathbf{I}) \quad (4)$$

where $\epsilon_\theta \sim \mathcal{N}(0, 1)$ represents a normal distribution noise with a mean of 0 and a variance of 1. Through induction, we can directly obtain $\mathbf{H}_t$ from $\mathbf{H}_0$ by

$$\mathbf{H}_t = \sqrt{\bar{\alpha}_t}\mathbf{H}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon_\theta, \epsilon_\theta \sim \mathcal{N}(0, 1) \quad (5)$$

$$\bar{\alpha}_t = \prod_{j=1}^t \alpha_j \quad (6)$$

where $\bar{\alpha}_t$ represents the cumulative product of the retention ratio α_t up to time step t . To preserve the structural information of the input HSI, we extracted the low-frequency guided feature $\mathbf{H}'_{0,t}$ of the input HSI $\mathbf{H}_0$ and the low-frequency guided feature $\mathbf{H}'_{t,t}$ of the noisy HSI $\mathbf{H}_t$ at each time step through the LFGM. Then, the two low-frequency guided features are fused with the corresponding noise HSI. The fusion process is defined as

$$\mathbf{H}''_t = \mathbf{H}_t \otimes \mathbf{H}'_{0,t} \otimes \mathbf{H}'_{t,t} \quad (7)$$

where $\otimes$ represents Hadamard product, $\mathbf{H}''_t$ represents the processed HSI. The processed HSIs, including $\mathbf{H}_0, \mathbf{H}''_1, \mathbf{H}''_2, \dots, \mathbf{H}''_t, \dots, \mathbf{H}''_T$, are used as training data. It is worth noting that the processed training sample $\mathbf{H}''_t$ obtained by (7) does not follow the original forward diffusion distribution $q(\mathbf{H}_t|\mathbf{H}_0)$, because the $\mathbf{H}''_t$ is obtained by the explicit combination of $\mathbf{H}_t$ with $\mathbf{H}'_{0,t}$ and $\mathbf{H}'_{t,t}$ through Hadamard product. This introduces a deterministic structural bias that deviates from the original forward diffusion process. This design imposes a global structure prior on the noise input $\mathbf{H}_t$ to ensure that the diffusion process will not completely erase the global spatial layout of the HSI during training. In addition, low-frequency guided features usually correspond to large-scale background structures in the HSI. By introducing these features, we guided the network to learn the noise distribution while maintaining its perception of the overall spatial structure of the input HSI. This can alleviate the problem of image structure degradation in the standard diffusion process, thereby achieving stable training and maintaining the coherence of the background structure in the generated HSI. Finally, through the introduction of low-frequency guided features, the stability of network training will not be destroyed.

To guide the network to accurately generate HSI and anomaly targets, we extracted the sampling guided feature $\mathbf{A}$ of the input HSI through HAFGM. Then, the extracted sampling guided feature $\mathbf{A}$ is embedded into the reverse diffusion process. The reverse diffusion process with embedded sampling guided feature $\mathbf{A}$ is defined as

$$p_\theta(\mathbf{Z}_{0:T}) = p(\mathbf{Z}_T) \prod_{t=1}^T p_\theta(\mathbf{Z}_{t-1}|\mathbf{Z}_t, \mathbf{A}) \quad (8)$$

$$p_\theta(\mathbf{Z}_{t-1}|\mathbf{Z}_t, \mathbf{A}) = \mathcal{N}(\mathbf{Z}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{Z}_t, \mathbf{A}, t), \sigma^2_t \mathbf{I}) \quad (9)$$

where $\mathbf{Z}_T$ represents the noise HSI that follows a standard Gaussian distribution, $\mathbf{Z}_0$ represents the generated HSI, $\mathbf{Z}_t$ represents the estimated image at the t -th time step, $p_\theta(\cdot)$ represents the reverse diffusion process, θ represents the parameters of the model, $\boldsymbol{\mu}_\theta(\mathbf{Z}_t, \mathbf{A}, t)$ and $\sigma^2_t \mathbf{I}$ represent the mean and variance during the reverse diffusion process, respectively. In the proposed model, inspired by the conditional diffusion framework, we designed a novel reverse diffusion process. The reverse diffusion process can be computed as

$$\mathbf{Z}_{t-1} = \bar{\boldsymbol{\mu}}_\theta(\mathbf{Z}_t, \mathbf{A}, t) + w(t)\mathbf{A} + \sqrt{\beta_t}\varepsilon_\theta, \varepsilon_\theta \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (10)$$

where

$$\bar{\boldsymbol{\mu}}_\theta(\mathbf{Z}_t, \mathbf{A}, t) = \frac{1}{\sqrt{\alpha_t}} \left[\mathbf{Z}_t - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} \varepsilon_\theta(\mathbf{Z}_t, \mathbf{A}, t) \right] \quad (11)$$

$$w(t) = \frac{\sqrt{1 - \alpha_t} + \sqrt{\alpha_t}}{\sqrt{\alpha_t(1 - \alpha_t)}} \quad (12)$$

where $\varepsilon_\theta(\mathbf{Z}_t, \mathbf{A}, t)$ represents the estimated noise at the current time step t , $\bar{\boldsymbol{\mu}}_\theta(\mathbf{Z}_t, \mathbf{A}, t)$ is the mean in the standard conditional diffusion, $w(t)$ is the weight coefficient related to the time step. The estimated image $\mathbf{Z}_t$ at the t -th time step and the sampling guided feature $\mathbf{A}$ are inputs. The equation (10) adds an explicit guiding term via the additive term $w(t)\mathbf{A}$. After we added $w(t)\mathbf{A}$, the conditional transition distribution $p_\theta(\mathbf{Z}_{t-1}|\mathbf{Z}_t, \mathbf{A})$ is still Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{Z}_t, \mathbf{A}, t), \sigma^2_t \mathbf{I})$, where the mean term $\boldsymbol{\mu}_\theta(\mathbf{Z}_t, \mathbf{A}, t)$ is expressed as

$$\boldsymbol{\mu}_\theta(\mathbf{Z}_t, \mathbf{A}, t) = \bar{\boldsymbol{\mu}}_\theta(\mathbf{Z}_t, \mathbf{A}, t) + w(t)\mathbf{A} \quad (13)$$

Hence, after adding the term $w(t)\mathbf{A}$, the mean is shifted by $w(t)\mathbf{A}$ while the variance keeps unchanged. This revision is consistent with the diffusion framework. We obtained the generated HSI $\mathbf{Z}_0$ by iteratively executing the reverse diffusion process. For the generated HSI $\mathbf{Z}_0$, it is not intended to perfectly reconstruct the input HSI $\mathbf{H}_0$, but rather serves as a structure guided, edge, and anomaly feature fused generated image. The generated HSI $\mathbf{Z}_0$ preserves the global structural information of the background while integrating high-frequency edge details and anomaly target features. The input HSI $\mathbf{H}_0$ contains the real background and real anomaly targets. The generated HSI $\mathbf{Z}_0$ contains the reconstructed background and synthesized anomaly targets. These synthetic anomaly targets are generated based on prior guided knowledge, rather than simply replicating real anomaly targets. The generated HSI $\mathbf{Z}_0$ is an image designed to have controllable differences from the input HSI $\mathbf{H}_0$. This difference is used in subsequent fusion steps to extract anomaly targets.

Due to the presence of anomaly targets in the generated HSI $\mathbf{Z}_0$, we extracted anomaly targets by fusing the input HSI $\mathbf{H}_0$ with the generated HSI $\mathbf{Z}_0$. The fused HSI is defined as

$$\mathbf{F} = \mathbf{H}_0 + \mathbf{Z}_0 \quad (14)$$

where $\mathbf{F}$ is the fused HSI. Anomaly detection results can be extracted by the dot product of each spectrum in the fused HSI. Therefore, the anomaly detection result is written as

$$\mathbf{T}(m, n) = \sum_{l=1}^L \mathbf{F}(m, n, l)^2, m = 1, 2, \dots, M, n = 1, 2, \dots, N \quad (15)$$

where $\mathbf{T}(m, n)$ represents the value at the spatial position (m, n) in the anomaly detection result $\mathbf{T}$, $\mathbf{F}(m, n, l)$ represents the value at position (m, n) in the l -th band of the fused image $\mathbf{F}$, and L represents the number of bands in the HSI. In the anomaly detection result $\mathbf{T}$, the larger the value

of $\mathbf{T}(m, n)$, the higher the probability that the pixel is an anomaly target.

3.2. LFGM

The LFGM was proposed to extract low-frequency guided feature from both the input HSI and the noise HSI. This LFGM employs a dynamic threshold segmentation WT strategy, which suppresses noise while preserving global structural feature in HSIs.

Traditional filter methods are suitable for stationary signals or simple scenarios that require high real-time performance. The WT has multi-scale analysis capability and adaptive time-frequency characteristic, making it suitable for processing local features or non-stationary signals [67]. The WT demonstrates superior capability in processing complex background. Therefore, in the proposed LFGM, WT was used to perform multi-level decomposition on the HSI. After WT, dynamic threshold segmentation is performed on high-frequency detail components to remove some noise while preserving the global structural information of the input HSI. The overall flowchart of the LFGM is shown in Fig. 2.

We assume that the input image of WT is represented as $\mathbf{X} \in \mathbf{R}^{M \times N \times L}$, and the wavelet basis of WT is Daubechies4 wavelet. In the WT, the scale function coefficient and wavelet function coefficient are represented through low-pass filter kernel and high-pass filter kernel. Therefore, the essence of WT for input image is to perform row and column convolution operations on the image. The low-pass filter kernel and high-pass filter kernel corresponding to Daubechies4 wavelet are

$$\mathbf{K}_L = \left[\frac{1 + \sqrt{3}}{4\sqrt{2}}, \frac{3 + \sqrt{3}}{4\sqrt{2}}, \frac{3 - \sqrt{3}}{4\sqrt{2}}, \frac{1 - \sqrt{3}}{4\sqrt{2}} \right] \quad (16)$$

$$\mathbf{K}_H = \left[\frac{1 - \sqrt{3}}{4\sqrt{2}}, -\frac{3 - \sqrt{3}}{4\sqrt{2}}, \frac{3 + \sqrt{3}}{4\sqrt{2}}, -\frac{1 + \sqrt{3}}{4\sqrt{2}} \right] \quad (17)$$

where $\mathbf{K}_L$ and $\mathbf{K}_H$ represent low-pass filter kernel and high-pass filter kernel, respectively. The specific steps for performing a first level WT on the input image $\mathbf{X}$ are as follows. First, we performed row-wise convolution on each spectral band of image $\mathbf{X}$ using low-pass and high-pass filter kernels. Subsequently, column-wise convolution was applied to the row convolution results using low-pass and high-pass filter kernels, respectively. It is worth noting that the stride of row convolution and column convolution is 2. The row convolution is represented as

$$[\mathbf{I}_{\text{row}l}(:, :, l), \mathbf{I}_{\text{row}h}(:, :, l)] = C_{\text{onv}2}([\mathbf{K}_L, \mathbf{K}_H], \mathbf{X}(:, :, l)) \quad (18)$$

where $\mathbf{X}(:, :, l)$ is the l -th band of image $\mathbf{X}$, $C_{\text{onv}2}$ is a convolution operation with a stride of 2, $\mathbf{I}_{\text{row}l}(:, :, l)$ is the convolution result of the low-pass filter kernel $\mathbf{K}_L$ and the image $\mathbf{X}(:, :, l)$, $\mathbf{I}_{\text{row}h}(:, :, l)$ is the convolution result of the high-pass filter kernel $\mathbf{K}_H$ and the image $\mathbf{X}(:, :, l)$. The column convolution is represented as

$$[\mathbf{X}_{LL}^1(:, :, l), \mathbf{X}_{LH}^1(:, :, l)] = C_{\text{onv}2}([\mathbf{K}_L^T, \mathbf{K}_H^T], \mathbf{I}_{\text{row}l}(:, :, l)) \quad (19)$$

$$[\mathbf{X}_{HL}^1(:, :, l), \mathbf{X}_{HH}^1(:, :, l)] = C_{\text{onv}2}([\mathbf{K}_L^T, \mathbf{K}_H^T], \mathbf{I}_{\text{row}h}(:, :, l)) \quad (20)$$

where $\mathbf{X}_{LL}^1 \in \mathbf{R}^{M/2 \times N/2 \times L}$ is the low-frequency approximation component of the first level WT, $\mathbf{K}_L^T$ and $\mathbf{K}_H^T$ are the transposes of low-pass and high-pass filter kernels, $\mathbf{X}_{LH}^1 \in \mathbf{R}^{M/2 \times N/2 \times L}$, $\mathbf{X}_{HL}^1 \in \mathbf{R}^{M/2 \times N/2 \times L}$, and $\mathbf{X}_{HH}^1 \in \mathbf{R}^{M/2 \times N/2 \times L}$ are the high-frequency detail components of the first level WT in the horizontal, vertical, and diagonal directions, respectively. After obtaining the results of the first level WT, combined with (18)-(20), other level WT are represented as

$$[\mathbf{X}_{LL}^i, \mathbf{X}_{LH}^i, \mathbf{X}_{HL}^i, \mathbf{X}_{HH}^i] = W_T(\mathbf{X}_{LL}^{i-1}), i = 2, 3, 4 \quad (21)$$

where $W_T(\cdot)$ represents WT operation, i is the current level of WT, $\mathbf{X}_{i-1} LL$ and $\mathbf{X}_i LL$ are the low-frequency approximation components of the $i-1$ -th and i -th level WT, respectively, $\mathbf{X}_i LH$, $\mathbf{X}_i HL$, and $\mathbf{X}_i HH$ are the high-

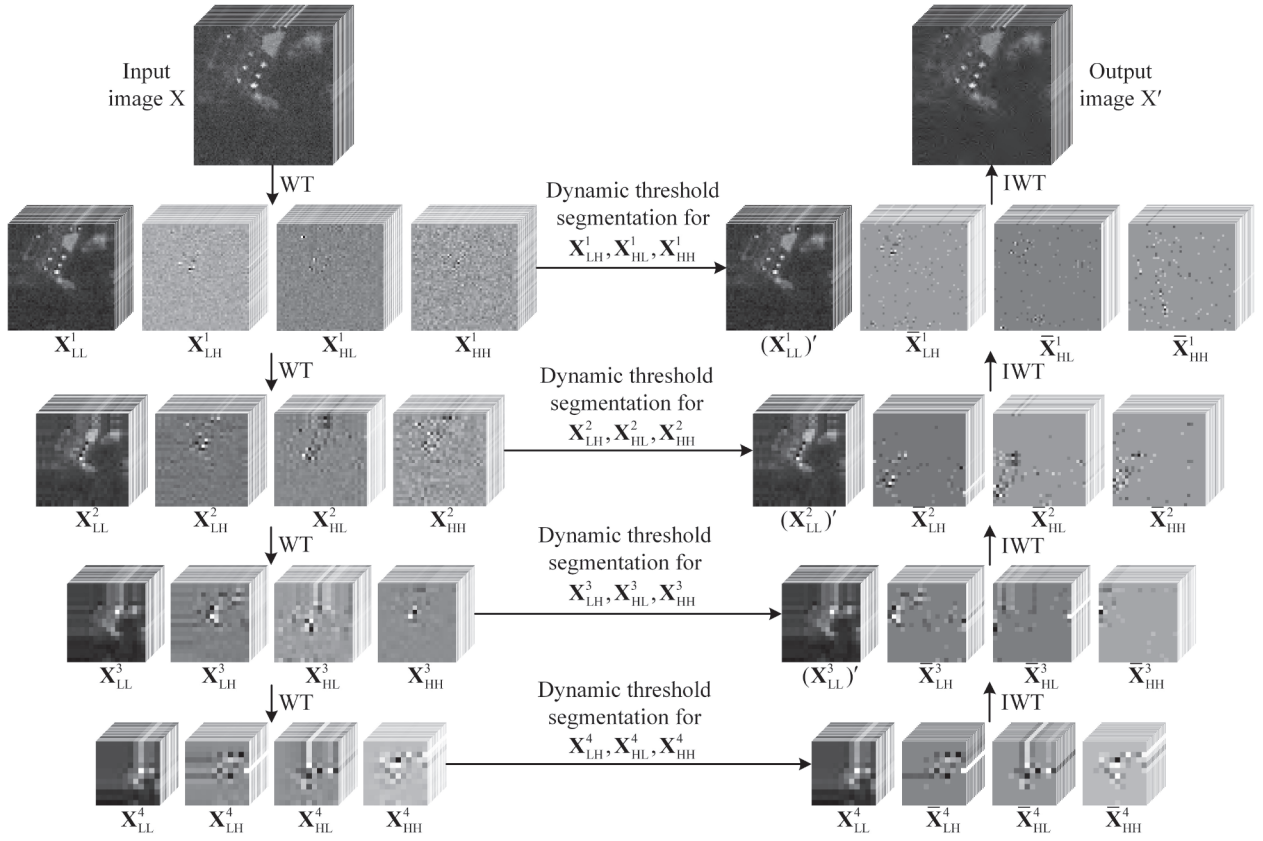


Fig. 2. Overall flowchart of the proposed LFGM.

frequency detail components of the i -th level WT in the horizontal, vertical, and diagonal directions, respectively. Through the above operation, we can achieve a 4-level WT on the input image X .

To suppress some noise and preserve the global structural information of the input image, we implemented threshold segmentation on the high-frequency detail components of each level of WT. For the threshold, inspired by the VisuShrink threshold [68] and combined with the forward diffusion process, we proposed a dynamic threshold strategy. By introducing the added noise intensity parameter β_t , we added a dynamic threshold factor to the VisuShrink threshold to automatically adjust the threshold. The dynamic threshold is represented as

$$y_t(l, k) = \frac{1}{1 + e^{-\beta_t}} \times \frac{m_{ad}(l)}{0.6745} \times \sqrt{2 \ln \left(\frac{M}{2k} \times \frac{N}{2k} \right)} \quad (22)$$

Where $m_{ad}(l)$ is the median of the l -th band in the high-frequency detail component, $k \in [1, 4]$ is the WT level of the current calculated high-frequency detail component, $y_t(l, k)$ is the threshold for the k -th level high-frequency detail in the l -th band of the image with added noise at the t -th time step. Since noise primarily concentrates in high-frequency detail components and exhibits small absolute values. As such, the threshold segmentation of high-frequency detail components is represented as

$$\bar{X}_{LH}^k(:, :, l) = \begin{cases} X_{LH}^k(:, :, l), & \text{if } |X_{LH}^k(:, :, l)| \geq y_t(l, k) \\ 0, & \text{if } |X_{LH}^k(:, :, l)| < y_t(l, k) \end{cases} \quad (23)$$

$$\bar{X}_{HL}^k(:, :, l) = \begin{cases} X_{HL}^k(:, :, l), & \text{if } |X_{HL}^k(:, :, l)| \geq y_t(l, k) \\ 0, & \text{if } |X_{HL}^k(:, :, l)| < y_t(l, k) \end{cases} \quad (24)$$

$$\bar{X}_{HH}^k(:, :, l) = \begin{cases} X_{HH}^k(:, :, l), & \text{if } |X_{HH}^k(:, :, l)| \geq y_t(l, k) \\ 0, & \text{if } |X_{HH}^k(:, :, l)| < y_t(l, k) \end{cases} \quad (25)$$

where $X_{LH}^k(:, :, l)$, $X_{HL}^k(:, :, l)$ and $X_{HH}^k(:, :, l)$ represent the l -th band of the high-frequency detail component of the k -th level WT, $\bar{X}_{LH}^k(:, :, l)$, $\bar{X}_{HL}^k(:, :, l)$ and $\bar{X}_{HH}^k(:, :, l)$ represent the l -th band of the high-frequency detail component of the k -th level WT after threshold segmentation. After threshold segmentation of all high-frequency detail components are completed, the inverse wavelet transform is used to obtain the output image by

$$X' = \text{IWT}(X_{LL}^4, \bar{X}_{LH}^4, \bar{X}_{HL}^4, \bar{X}_{HH}^4, \bar{X}_{LH}^3, \bar{X}_{HL}^3, \bar{X}_{HH}^3, \bar{X}_{LH}^2, \bar{X}_{HL}^2, \bar{X}_{HH}^2, \bar{X}_{LH}^1, \bar{X}_{HL}^1, \bar{X}_{HH}^1) \quad (26)$$

where $\text{IWT}(\cdot)$ represents the inverse wavelet transform operation, X' is the output image of the LFGM, which is also a low-frequency guided feature. The detailed implementation steps of the LFGM are shown in Algorithm 1.

Algorithm 1 LFGM

Input: HSI X and intensity parameters of noise β_t

For each band $X(:, :, l)$ **do**

1: Implement first level WT by (16)-(20);

2: Implement other level WT by (21);

3: Calculate the threshold corresponding to each level of WT by (22);

4: Perform threshold segmentation on high-frequency detail components by (23)-(25);

5: Reconstruct image X' through inverse WT according to (26);

Output: Low-frequency guided feature

3.3. HAFGM

We constructed the HAFGM to guide the network to accurately generate the HSI and anomaly targets. The overall structure of the HAFGM is shown in Fig. 3. To extract high-frequency edge features of the HSI $\mathbf{H}_0$, we adopted the Sobel filter method. First, all bands in the HSI $\mathbf{H}_0$ are fused into one band. The fusion result is

$$\mathbf{d} = \sum_{l=1}^L \mathbf{H}_0(:, :, l) \quad (27)$$

where $\mathbf{d}$ is the fused image of all bands, and $\mathbf{H}_0(:, :, l)$ is the l -th band in the HSI $\mathbf{H}_0$. Then, we assumed that the Sobel filter kernels in the horizontal and vertical directions are represented as $\mathbf{g}_h$ and $\mathbf{g}_v$, respectively. The filter result is

$$\mathbf{D}_h = C_{ov}(\mathbf{d}, \mathbf{g}_h) \quad (28)$$

$$\mathbf{D}_v = C_{ov}(\mathbf{d}, \mathbf{g}_v) \quad (29)$$

$$\mathbf{B}_{or} = \sqrt{\mathbf{D}_h^2 + \mathbf{D}_v^2} \quad (30)$$

where $C_{ov}(\cdot)$ represents a convolution operation with a stride of 1, $\mathbf{D}_h$ and $\mathbf{D}_v$ represent the horizontal and vertical filter results of the fused image $\mathbf{d}$, respectively, and $\mathbf{B}_{or}$ is the filtered output result and also the edge feature of the HSI.

For the anomaly features, we adopted a spectral correlation method. Since background pixels outnumber anomaly pixels in the HSI, the correlations between background pixels are prioritized. To reduce the contamination of the background by anomaly targets, dual window strategy is employed to traverse the HSI. The spectral correlation between the tested pixel and each pixel of the outer window can be calculated by

$$e(m', n') = \frac{\sum_{l=1}^L (\mathbf{h}_{m,n} - \bar{h}) \odot (\mathbf{p}_{m',n'} - \bar{p})}{\sqrt{\sum_{l=1}^L (\mathbf{h}_{m,n} - \bar{h})^2 \sum_{l=1}^L (\mathbf{p}_{m',n'} - \bar{p})^2}} \quad (31)$$

where $\mathbf{h}_{m,n}$ represents the tested pixel, $\bar{h}$ represents the mean of the tested pixel $\mathbf{h}_{m,n}$, $\odot$ represents dot product operation, $\mathbf{p}_{m',n'}$ represents the pixels located between the inner and outer windows in the dual window, $\bar{p}$ represents the mean of pixel $\mathbf{p}_{m',n'}$, and $e(m', n')$ represents the spectral correlation between the tested pixel and each pixel of the outer window. After the spectral correlations between the tested pixel and all pixels in the outer window were calculated, we selected the maximum spectral correlation value as the final spectral correlation of the tested pixel. The maximum spectral correlation value is written as

$$\mathbf{B}_{sc}(m, n) = \max(e(m', n')) \quad (32)$$

where $\mathbf{B}_{sc}(m, n)$ represents the value at (m, n) in the spectral correlation

image $\mathbf{B}_{sc}$, $\max(\cdot)$ represents the operation of selecting the maximum value. The spectral correlation image $\mathbf{B}_{sc}$ was obtained by traversing all pixels in the HSI $\mathbf{H}_0$ using the above method. Since $\mathbf{B}_{sc}$ represents the spectral correlation between background pixels, we applied a reverse operation on $\mathbf{B}_{sc}$ to obtain anomaly features. The reverse operation is expressed as

$$\mathbf{B}_{am} = e^{\ln(-\mathbf{B}_{sc}+2)} \quad (33)$$

where $\mathbf{B}_{am}$ represents anomaly feature. In (33), since the value of $\mathbf{B}_{sc}$ belongs to $[-1, 1]$, we added 2 to the logarithmic function to ensure that the result is a positive number. To further increase the difference between anomaly and background, an exponential operation is performed on the logarithmic result. After obtaining the edge feature $\mathbf{B}_{or}$ and anomaly feature $\mathbf{B}_{am}$, the final sampling guided feature is represented as

$$\mathbf{A} = \hat{h}(\mathbf{B}_{or} \otimes \mathbf{B}_{am}) \quad (34)$$

where $\hat{h}(\cdot)$ represents the normalization operation of maximum and minimum values, $\mathbf{A}$ is the output of the HAFGM, which is the sampling guided feature.

3.4. Model training

In the proposed model, we selected a multi-layer AE network with fully connected layers as the backbone network to train the proposed diffusion model. The structure of the network is shown in Fig. 4. After each fully connected layer, the Sigmoid function is used as the activation function. During the training process, we used each band of the HSI as input for training. During the generation process, we added a total variation filter at the intermediate time step $T/2$ to further remove noise. The loss function consists of mean squared error (MSE) loss, Kullback Leibler (KL) divergence loss, and total variation regularization loss. The MSE loss is

$$L_{oss1} = \frac{1}{M \times N} \sum_{m=1}^M \sum_{n=1}^N [\mathbf{W}_{tru}(m, n) - \varepsilon_{\theta}(\mathbf{Z}_t, \mathbf{A}, t)(m, n)]^2 \quad (35)$$

where $\mathbf{W}_{tru} \sim \mathcal{N}(0, \mathbf{I})$ represents real noise sampled from the standard

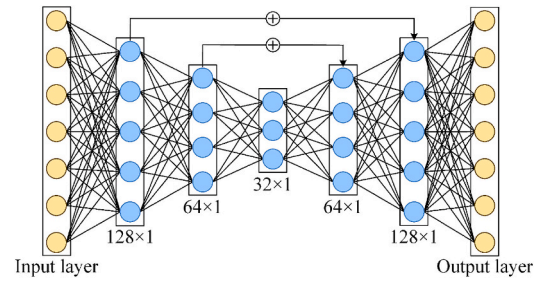


Fig. 4. Network architecture of the multi-layer AE.

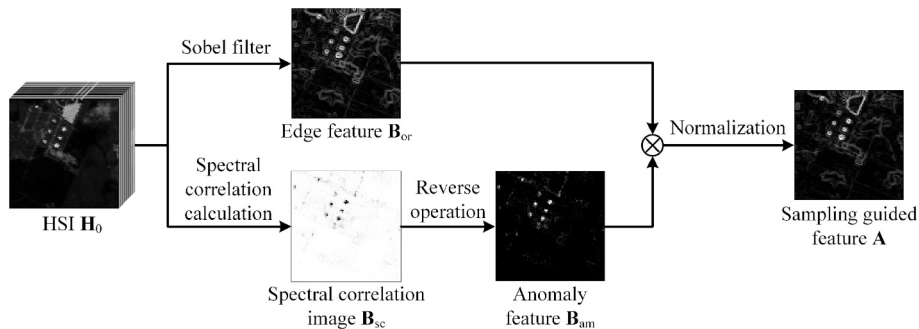


Fig. 3. Architecture of the proposed HAFGM.

Gaussian distribution. The KL divergence loss is

$$L_{\text{oss2}} = \frac{1}{2 \times M \times N} \sum_{m=1}^M \sum_{n=1}^N \left[\frac{\mathbf{W}_{\text{tru}}^2(m, n)}{T^2 + \delta} - \ln \left(\frac{\mathbf{W}_{\text{tru}}^2(m, n)}{T^2 + \delta} \right) - 1 \right] \quad (36)$$

where δ is a minimum value used to avoid numerical instability. The total variation regularization loss is

$$L_{\text{oss3}} = \frac{1}{M \times N \times L} \sum_{m=1}^M \sum_{n=1}^N \sum_{l=1}^L |\nabla_z \mathbf{W}_{\text{pre}}(m, n, l)| \quad (37)$$

where $\nabla_z \mathbf{W}_{\text{pre}}(m, n, l)$ represents the spectral gradient at spatial position (m, n) and l -th band. The total loss function is

$$L_{\text{total}} = L_{\text{oss1}} + \lambda_1 L_{\text{oss2}} + \lambda_2 L_{\text{oss3}} \quad (38)$$

where λ_1 and λ_2 are penalty coefficients. The detailed implementation of the proposed high-low-frequency feature-guided diffusion model is shown in Algorithm 2.

Algorithm 2 Proposed high-low-frequency feature-guided diffusion model

Input: HSI $\mathbf{H}_0$

Initialization: Outer window size W_{out} , inner window size W_{in} , penalty coefficient λ_1 and λ_2

- 1: Add noise to input HSI $\mathbf{H}_0$ according to (5) and (6);
- 2: Calculate low-frequency guided features $\mathbf{H}_l^*$ and $\mathbf{H}_{0,l}^*$ according to Algorithm 1;
- 3: Obtain new training data $\mathbf{H}^*$ by (7);
- 4: Train the proposed model by the backpropagation method by combining (35)-(38);
- 5: Calculate edge feature $\mathbf{B}_{\text{or}}$ according to (27)-(30);
- 6: Calculate anomaly feature $\mathbf{B}_{\text{am}}$ according to (31)-(33);
- 7: Calculate sampling guided feature $\mathbf{A}$ by (34);
- 8: Generate HSI $\mathbf{Z}_0$ by iteratively executing the reverse diffusion process;
- 9: Extract anomaly targets by (14) and (15);

Output: Anomaly detection result $\mathbf{T}$

4. Experimental results and analysis

In this section, we first introduced the experimental dataset, evaluation metric, and state-of-the-art compared methods. Second, the proposed method is compared with the results of other state-of-the-art compared methods to verify the superiority. Finally, we analyzed the framework performance of the proposed method through ablation experiment. All experiments were conducted in MATLAB 2024b and Python 3 on a computer with an Intel Core i5-12400 CPU and 32-GB RAM. The operating system of the computer is Windows 10 professional. The proposed method is denoted as HFGDM.

4.1. Experimental dataset

- (1) *Texas Coast I and II*: The two datasets were collected using the airborne visible/infrared imaging spectrometer (AVIRIS) sensor on the Texas Coast in the United States [69]. The two datasets have 100×100 pixels. The Texas Coast I dataset contains 204 bands, while the Texas Coast II dataset has 207 bands. The spatial resolution of both datasets is 17.2 m per pixel. The pseudo color images and ground-truth images of the two datasets are shown in Fig. 5(a) and (b).
- (2) *Los Angeles*: This dataset was collected using the AVIRIS sensor in Los Angeles, California, USA. The spatial size of this dataset is 100×100 and the spectral bands of this dataset are 205. Its spatial resolution is 7.1 m per pixel. The pseudo color image and ground-truth image of this dataset are shown in Fig. 5(c).
- (3) *Bay Champagne*: This dataset was collected using AVIRIS sensors in Bay Champagne. It has a total of 100×100 pixels and 188 bands. Its spatial resolution is 4.4 m per pixel. The pseudo color image and ground-truth image of this dataset are shown in Fig. 5(d).
- (4) *Pavia*: This dataset was collected using the reflective optics system imaging spectrometer (ROSIS-03) sensor in Pavia, Italy. In the experiment, a total of 100×100 areas were cropped for experiment. This dataset has a total of 102 bands with a spatial resolution of 1.3 m per pixel. The pseudo color image and ground-truth image of this dataset are shown in Fig. 5(e).

4.2. Evaluation metric

In the experiment, the three-dimensional receiver operating characteristic (3-D ROC) curve [70–71], the corresponding two-dimensional ROC (2-D ROC) curve, and the area under the 2-D ROC curve (AUC) used as an evaluation metric. The 3-D ROC is represented as (P_D, P_F, τ) , where P_D represents detection probability, P_F represents false alarm probability, and τ represents threshold. Three 2-D ROC curves are represented as (P_D, P_F) , (P_D, τ) and (P_F, τ) , respectively, with corresponding AUC values represented as $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$, and $\text{AUC}_{(F,\tau)}$. Four other AUC values are calculated based on these three AUC values. These four AUC values are

$$\text{AUC}_{\text{TD}} = \text{AUC}_{(D,F)} + \text{AUC}_{(D,\tau)} \quad (39)$$

$$\text{AUC}_{\text{BS}} = \text{AUC}_{(D,F)} - \text{AUC}_{(F,\tau)} \quad (40)$$

$$\text{AUC}_{\text{TDBS}} = \text{AUC}_{(D,\tau)} - \text{AUC}_{(F,\tau)} \quad (41)$$

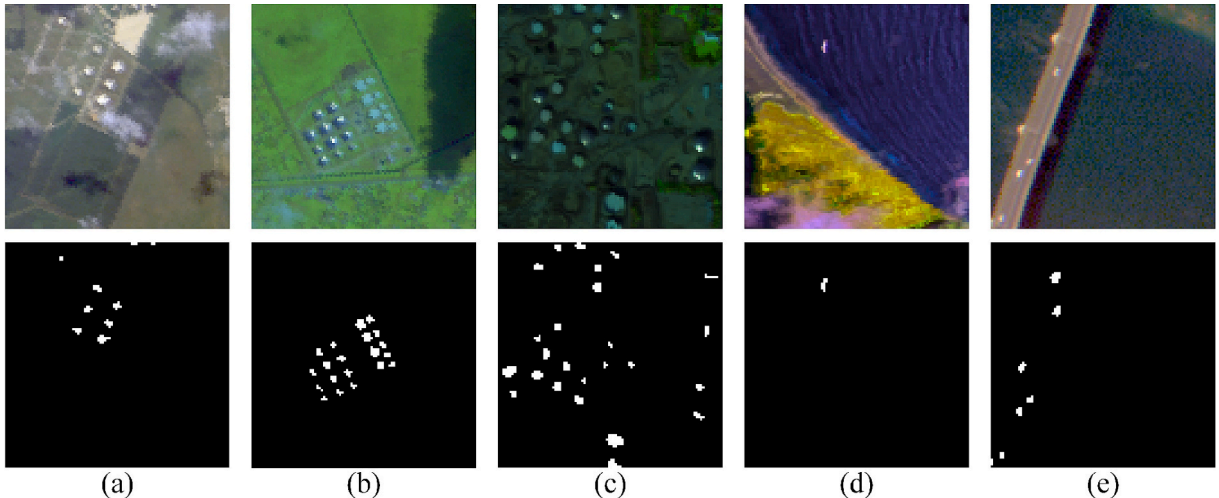


Fig. 5. Pseudo color image and ground-truth image of experimental datasets. (a) Texas Coast I. (b) Texas Coast II. (c) Los Angeles. (d) Bay Champagne. (e) Pavia.

$$AUC_{ODP} = AUC_{(D,F)} + AUC_{(D,\tau)} - AUC_{(F,\tau)} \quad (42)$$

$AUC_{(D,\tau)}$ and AUC_{TD} are used to evaluate the performance of anomaly detection. $AUC_{(F,\tau)}$ and AUC_{BS} are used to evaluate the performance of background suppression. $AUC_{(D,F)}$, AUC_{TDBS} , and AUC_{ODP} are used to evaluate overall performance. Among the seven AUC values, the smaller the AUC_{BS} value, the better the background suppression performance. The larger the other AUC values, the better the corresponding performance.

4.3. State-of-the-art compared method

In this study, the used state-of-the-art compared methods including the GRX [21], LEBSR [40], GAED [49], Auto-AD [48], RGAE [47], BockNet [53], DFAN [52], BSDM [19], and NL2Net [54] methods. The GRX, LEBSR, GAED, RGAE, and the proposed HFGDM methods were implemented on MATLAB 2024b. The remaining methods were run on Python 3. All parameters in the compared methods are set based on the parameters provided by the original authors or $AUC_{(D,F)}$ value. The GRX and Auto-AD methods do not require setting additional parameters. For other compared methods, the parameter settings are shown in Table 1.

In the LEBSR method, the parameters contain three tradeoff parameters α , β , λ , the value p , and number of endmembers e . For the GAED method, the parameters include window size c , learning rate l_a , penalty coefficient β , and number of iterations r_{iter} . In the RGAE method, the parameters that need to be set include tradeoff parameter λ , number of superpixels S , and dimension of hidden layer n_{hid} . For the BockNet method, the parameter that needs to be set is the width of guard window B_s . In the DFAN method, the parameters that need to be set are background category M , abnormal category E , the number of epochs E_p , and learning rate L_r . For the BSDM method, the parameters that need to be set are inference count K and time step t . In the NL2Net method, the parameter is stride size F .

4.4. Parameter analysis

In the proposed HFGDM method, the wavelet decomposition level is 4. The layer dimensions of the AE network are 128, 64, 32, 64, and 128, respectively. The filter kernel size and regularization strength of the total variation filter are 3×3 and 0.2, respectively. The diffusion time steps T is 1000. The intensity parameter of the added noise $\beta_t \in [0.0001, 0.01]$. The learning rate is 0.0001. In the experiments, the proposed HFGDM method is trained in an unsupervised manner using the entire input HSI as the training data. On all datasets, we used the Adam optimizer and trained for 50 epochs with a mini-batch size of 10.

The uncertain parameters in the proposed HFGDM method include the outer window size W_{out} and inner window size W_{in} of the dual window, two penalty coefficients λ_1 and λ_2 , the levels of WT k . When analyzing one parameter, the other parameters were kept unchanged. In the analysis process, $AUC_{(D,F)}$ value is used as evaluation metrics. The default values for these parameters are $W_{out} = 7$, $W_{in} = 5$, $\lambda_1 = 0.01$, $\lambda_2 = 0.01$, and $k = 4$.

- (1) *Dual window size (W_{in} , W_{out}):* When we changed the size of the dual window, kept the two penalty coefficients unchanged. The

value of W_{in} was changed from 3 to 9, and the value of W_{out} was changed according to the value of W_{in} . The impact of different W_{in} and W_{out} values on the detection results is shown in Table 2. According to Table 2, the values of (W_{in} , W_{out}) on the Texas Coast I and Los Angeles datasets are set to (7, 9). On other datasets, the values of (W_{in} , W_{out}) are uniformly set to (5, 7).

- (2) *Penalty coefficients λ_1 and λ_2 :* When we analyzed a penalty coefficient, other parameters were set to default values. The values of both penalty coefficients were varied from 0.0001 to 1. The impact of different penalty coefficient values on the detection results is shown in Fig. 6. According to Fig. 6(a), the value of λ_1 is uniformly set to 0.01 on five datasets. According to Fig. 6(b), the value of λ_2 is uniformly set to 0.1 on five datasets.
- (3) *Level of WT k :* When we analyzed the level of the WT k , other parameters were set to default values. The value of k was varied from 2 to 5. The impact of different values of k on the detection results is shown in Fig. 6(c). According to Fig. 6(c), when the value of k is uniformly set to 4 on five datasets, the proposed method achieves the best detection performance.

4.5. Experimental result

The detection results of the proposed and compared methods on five datasets are shown in Fig. 7. As shown in Fig. 7(a), on the Texas Coast I dataset, all methods detected abnormal targets. However, the background suppression effects of the DFAN, the GRX, the LEBSR, and the GAED methods are not excellent. The detection results of the RGAE and the proposed HFGDM methods are similar, with the top small piece of background not well suppressed. Other methods suppress the background while detecting anomaly targets.

Fig. 7(b) shows the detection results of different anomaly detection methods on the Texas Coast II dataset. The detection results of the DFAN method have low discrimination between background and anomaly

Table 2

The impact of different window sizes on detection result on five datasets.

W_{in}	W_{out}	Dataset Texas Coast I	Texas Coast II	Los Angeles	Bay Champagne	Pavia
3	5	0.9950	0.9989	0.9958	0.9995	0.9961
	7	0.9950	0.9989	0.9956	0.9995	0.9967
	9	0.9951	0.9989	0.9958	0.9996	0.9966
5	7	0.9956	0.9995	0.9958	0.9998	0.9970
	9	0.9956	0.9995	0.9959	0.9997	0.9967
	11	0.9955	0.9993	0.9958	0.9997	0.9964
7	9	0.9957	0.9995	0.9960	0.9998	0.9961
	11	0.9957	0.9993	0.9960	0.9998	0.9962
	13	0.9956	0.9983	0.9959	0.9997	0.9965
9	11	0.9957	0.9993	0.9959	0.9997	0.9962
	13	0.9955	0.9983	0.9958	0.9996	0.9965
	15	0.9953	0.9983	0.9958	0.9997	0.9965

Table 1

Parameter settings of the state-of-the-art compared methods.

Method	Parameter	Texas Coast I	Texas Coast II	Los Angeles	Bay Champagne	Pavia
LEBSR	$\alpha, \beta, \lambda, p, e$	0.1,0.1,0.5,0.5,8	0.1,1.5,0.5,0.5,3	0.1,1.5,0.5,0.5,3	0.1,0.1,0.5,0.5,6	0.1,0.1,0.5,0.5,6
GAED	c, l_a, β, r_{iter}	3,0.1,1,300	3,0.2,1,300	7,0.4,1,400	5,0.2,1,300	7,0.3,1,300
RGAE	λ, S, n_{hid}	0.01,100,100	0.01,100,100	0.001,150,150	0.01,150,100	0.001,150,100
BockNet	B_s	5	5	9	7	15
DFAN	M, E, E_p, L_r	5,0.5,20,0.0001	5,0.5,20,0.0001	4,0.5,20,0.0001	3,0.5,20,0.0001	6,0.6,20,0.0001
BSDM	K, t	10,500	10,500	10,500	10,500	10,500
NL2Net	F	5	5	5	7	3

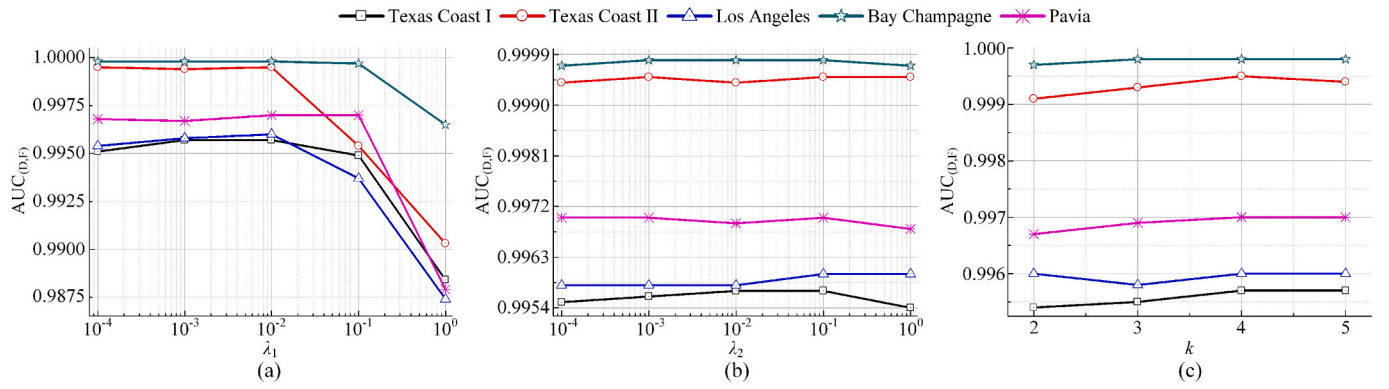


Fig. 6. Influence of two penalty coefficients and the level of the WT on detection performance on five datasets. (a) λ_1 . (b) λ_2 . (c) k .

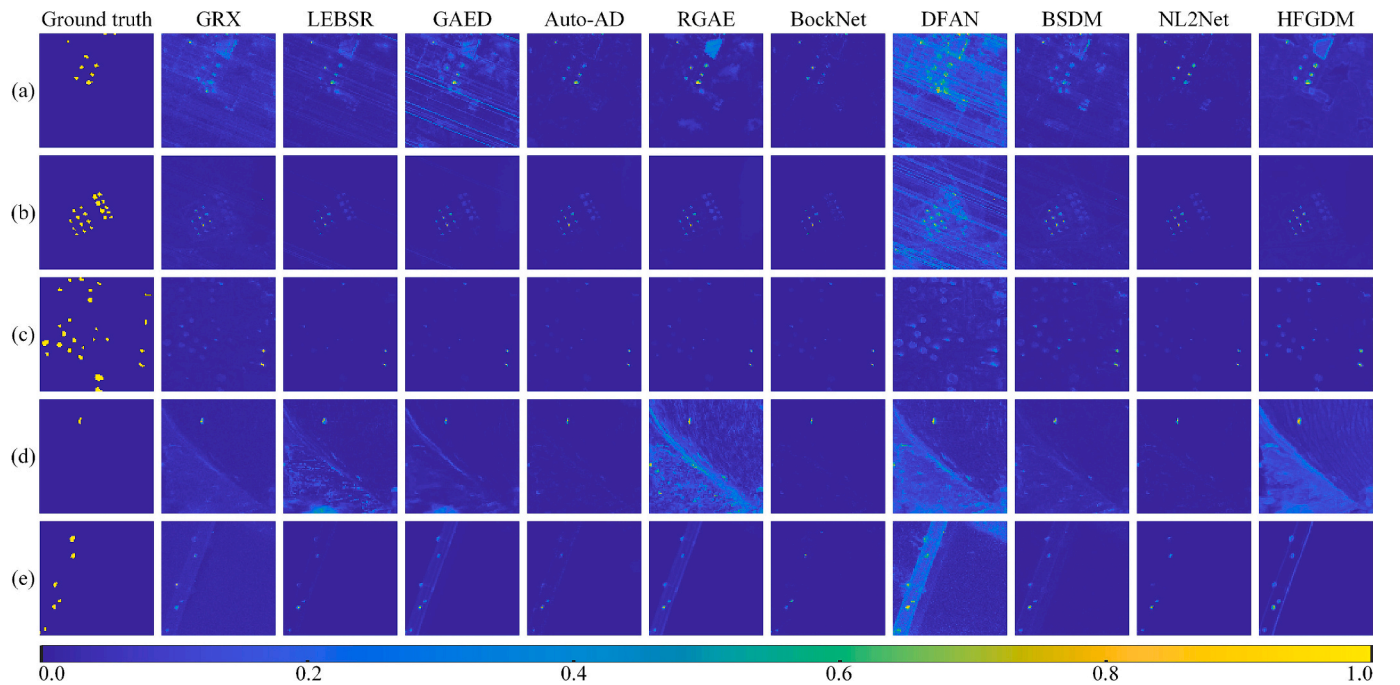


Fig. 7. Detection results of different HAD methods on five datasets. (a) Texas Coast I. (b) Texas Coast II. (c) Los Angeles. (d) Bay Champagne. (e) Pavia.

targets. The other compared methods and the proposed HFGDM method have achieved satisfactory detection results.

Fig. 7(c) shows the detection results of different anomaly detection methods on the Los Angeles dataset. In the detection results of the GRX, the DFAN, the BSDM, and the proposed HFGDM methods, anomaly targets are separated from the background. For the other compared methods, although anomaly targets are detected, the discrimination between anomaly targets and the background are relatively low.

As shown in Fig. 7(d), on the Bay Champagne dataset, the Auto-AD, the BockNet, and the NL2Net methods have achieved satisfactory detection results. The background suppression effect in the lower left corner of the GRX, the LEBSR, the GAED, and the BSDM methods is not particularly good. The detection results of the RGAE, the DFAN, and the proposed HFGDM methods are similar. The proposed HFGDM method did not suppress the background in the lower left corner, but the discrimination between anomaly targets and background is high.

As shown in Fig. 7(e), on the Pavia dataset, the detection result of the DFAN method show slightly poor background suppression. The suppression of bridge background in the detection results of the GRX, the GAED, and the RGAE methods is not very satisfactory. The background

suppression effect of other compared methods is satisfactory, but some anomaly targets are also suppressed. Although the proposed HFGDM method shows the outline of the bridge, anomaly targets have been successfully distinguished from the background.

The 3-D ROC and 2-D ROC curves of different anomaly detection methods are shown in Fig. 8. For the 2-D ROC curves (P_D , P_F), on five datasets, although the proposed HFGDM method has lower 2-D ROC curve (P_D , P_F) in some parts than some 2-D ROC curves (P_D , P_F). However, overall, the proposed HFGDM method has higher 2-D ROC curve (P_D , P_F) than other ROC curves (P_D , P_F). For the 2-D ROC curve (P_D , τ), the 2-D ROC curves (P_D , τ) of the DFAN method are higher than those of the other 2-D ROC curves on the first three datasets and the last dataset. On these four datasets, although the 2-D ROC curves (P_D , τ) of the proposed HFGDM method are not the highest, it is higher than the 2-D ROC curves (P_D , τ) of most compared methods. On the Bay Champagne dataset, the proposed HFGDM method obtained the optimal 2-D ROC curve (P_D , τ). For the 2-D ROC curve (P_F , τ), on the first two and fourth datasets, the 2-D ROC curves (P_F , τ) of the BockNet method are lower than those of other methods. On Los Angeles dataset, the LEBSR method obtained the lowest 2-D ROC curve (P_F , τ). On the last dataset,

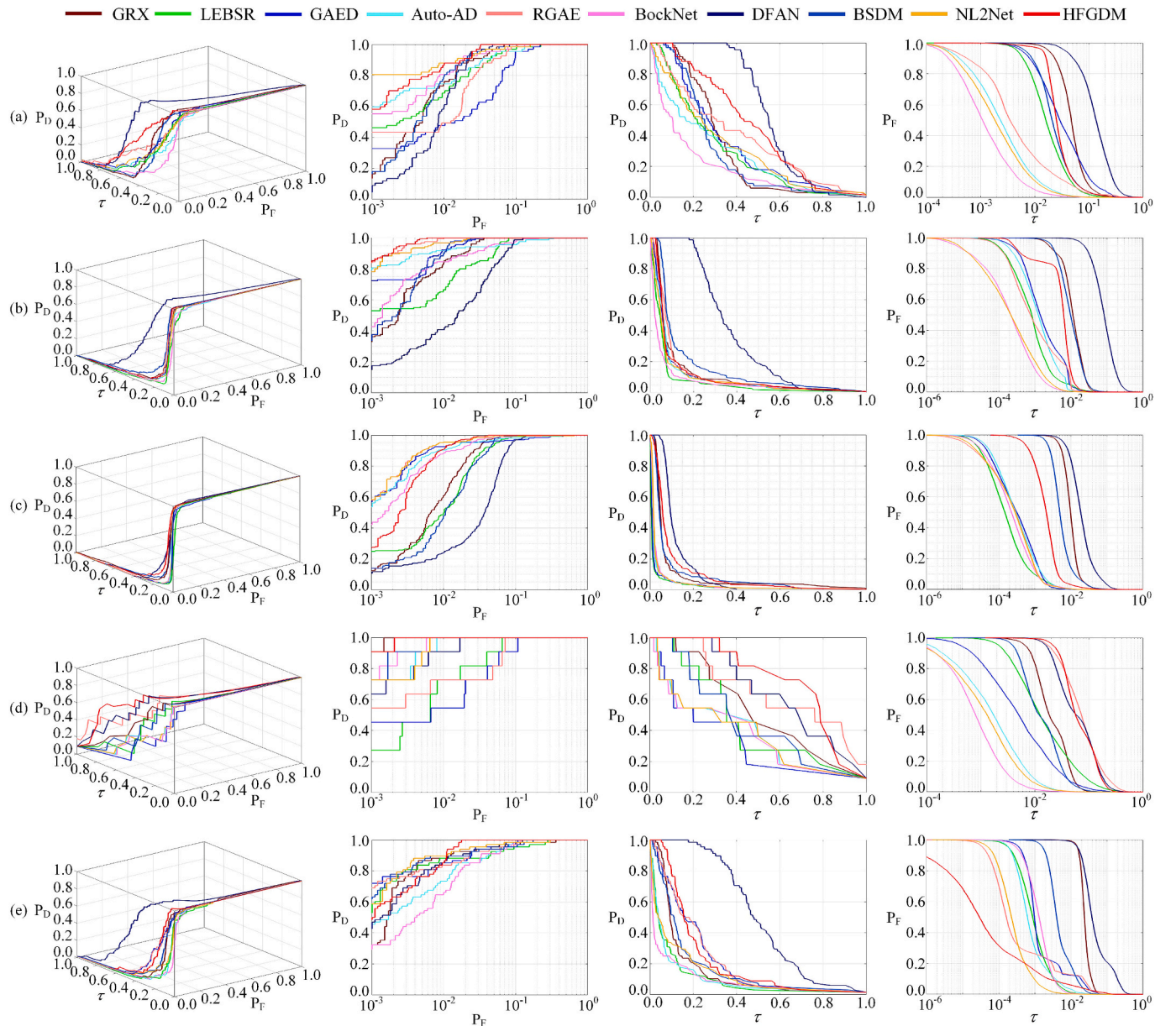


Fig. 8. Performance comparison of 3-D ROC and 2-D ROC curves of different HAD methods on five datasets. (a) Texas Coast I. (b) Texas Coast II. (c) Los Angeles. (d) Bay Champagne. (e) Pavia.

the NL2Net method obtained the lowest 2-D ROC curve (P_F , τ). The 2-D ROC curve (P_F , τ) of the proposed HFGDM method is not the highest on the five datasets.

The comparison of AUC values for different anomaly detection methods across five datasets is shown in Table 3. According to Table 3, for the Texas Coast I dataset, the proposed HFGDM method achieves peak performance in $AUC_{(D,F)}$, AUC_{TD} , AUC_{TDBS} , and AUC_{ODP} metrics. The DFAN method obtains the highest $AUC_{(D,\tau)}$ and AUC_{TD} values. The BockNet method obtains the minimum $AUC_{(F,\tau)}$ value. The NL2Net method obtains the maximum AUC_{BS} value. On the Texas Coast II dataset, the proposed HFGDM method obtains the maximum $AUC_{(D,F)}$ value. The DFAN method demonstrates superior performance in $AUC_{(D,\tau)}$, AUC_{TD} , AUC_{TDBS} , and AUC_{ODP} metrics. The BockNet method obtains the minimum $AUC_{(F,\tau)}$ value. The NL2Net method obtains the maximum AUC_{BS} value. For the Los Angeles dataset, the proposed HFGDM method achieves superior performance in $AUC_{(D,F)}$, AUC_{TD} , AUC_{TDBS} , and AUC_{ODP} metrics. The DFAN method obtains the maximum $AUC_{(D,\tau)}$ value. The LEBSR method obtains the minimum $AUC_{(F,\tau)}$ value. The RGAE and the

NL2Net methods achieve the same performance in AUC_{BS} metrics. For the Bay Champagne dataset, the proposed HFGDM method achieves best performance in $AUC_{(D,F)}$, AUC_{TD} , AUC_{TDBS} , and AUC_{ODP} metrics. The GRX method obtained the same $AUC_{(D,F)}$ value as the proposed HFGDM method. The BockNet method obtains the minimum $AUC_{(F,\tau)}$ value and the maximum AUC_{BS} value. For the Pavia dataset, the proposed HFGDM method achieves peak performance in both $AUC_{(D,F)}$ and AUC_{BS} metrics. The DFAN achieves peak performance in $AUC_{(D,\tau)}$, AUC_{TD} , AUC_{TDBS} , and AUC_{ODP} metrics. The NL2Net method obtains the minimum $AUC_{(F,\tau)}$ value. In summary, the proposed HFGDM method achieved satisfactory detection performance on five datasets.

4.6. Running time

To evaluate the computational efficiency of different anomaly detection methods, we compared the running time. In the experiment of this study, all programs were run using the CPU. The running time includes training time and testing time. The running time of different

Table 3

AUC values of different HAD methods on five datasets.

Dataset	AUC	Method GRX	LEBSR	GAED	Auto-AD	RGAE	BockNet	DFAN	BSDM	NL2Net	HFGDM
Texas Coast I	AUC _(D,F) ↑	0.9907	0.9854	0.9652	0.9876	0.9826	0.9917	0.9895	0.9914	0.9937	0.9957
	AUC _(D,r) ↑	0.3143	0.2876	0.3381	0.2695	0.3758	0.1886	0.5608	0.2860	0.3069	0.4560
	AUC _(F,r) ↓	0.0556	0.0238	0.0513	0.0054	0.0168	0.0029	0.1596	0.0303	0.0046	0.0339
	AUC _{TD} ↑	1.3049	1.2730	1.3033	1.2751	1.3585	1.1803	1.5504	1.2775	1.3006	1.4517
	AUC _{BS} ↑	0.9351	0.9616	0.9140	0.9822	0.9659	0.9888	0.8300	0.9612	0.9891	0.9618
	AUC _{TDBS} ↑	0.2587	0.2638	0.2868	0.2641	0.3591	0.1857	0.4013	0.2557	0.3023	0.4222
	AUC _{ODP} ↑	1.2494	1.2492	1.2521	1.2517	1.3417	1.1774	1.3908	1.2472	1.2960	1.4179
Texas Coast II	AUC _(D,F) ↑	0.9946	0.9875	0.9972	0.9905	0.9993	0.9910	0.9734	0.9964	0.9988	0.9995
	AUC _(D,r) ↑	0.1178	0.0724	0.1034	0.1003	0.1158	0.0712	0.3723	0.1579	0.0956	0.1110
	AUC _(F,r) ↓	0.0135	0.0020	0.0033	0.0022	0.0027	0.0006	0.1020	0.0128	0.0007	0.0065
	AUC _{TD} ↑	1.1124	1.0600	1.1007	1.0908	1.1151	1.0623	1.3457	1.1543	1.0944	1.1105
	AUC _{BS} ↑	0.9811	0.9855	0.9940	0.9883	0.9967	0.9904	0.8715	0.9836	0.9981	0.9930
	AUC _{TDBS} ↑	0.1043	0.0704	0.1002	0.0980	0.1131	0.0706	0.2703	0.1451	0.0949	0.1045
	AUC _{ODP} ↑	1.0989	1.0580	1.0974	1.0886	1.1124	1.0617	1.2438	1.1415	1.0937	1.1040
Los Angeles	AUC _(D,F) ↑	0.9987	0.9842	0.9891	0.9917	0.9948	0.9935	0.9616	0.9841	0.9948	0.9960
	AUC _(D,r) ↑	0.0891	0.0273	0.0313	0.0352	0.0391	0.0382	0.1215	0.0879	0.0387	0.1014
	AUC _(F,r) ↓	0.0114	0.0005	0.0006	0.0007	0.0006	0.0006	0.0272	0.0074	0.0006	0.0031
	AUC _{TD} ↑	1.0778	1.0115	1.0204	1.0269	1.0339	1.0317	1.0831	1.0720	1.0334	1.0974
	AUC _{BS} ↑	0.9773	0.9837	0.9885	0.9910	0.9942	0.9930	0.9344	0.9767	0.9942	0.9929
	AUC _{TDBS} ↑	0.0777	0.0268	0.0307	0.0345	0.0385	0.0376	0.0943	0.0804	0.0381	0.0984
	AUC _{ODP} ↑	1.0664	1.0110	1.0198	1.0262	1.0333	1.0311	1.0559	1.0646	1.0329	1.0944
Bay Champagne	AUC _(D,F) ↑	0.9998	0.9865	0.9770	0.9984	0.9823	0.9991	0.9974	0.9990	0.9985	0.9998
	AUC _(D,r) ↑	0.5314	0.4538	0.3251	0.3774	0.6923	0.3690	0.6595	0.4552	0.3757	0.7277
	AUC _(F,r) ↓	0.0259	0.0327	0.0148	0.0038	0.0987	0.0017	0.0838	0.0190	0.0031	0.0828
	AUC _{TD} ↑	1.5313	1.4403	1.3021	1.3758	1.6745	1.3681	1.6568	1.4542	1.3741	1.7275
	AUC _{BS} ↑	0.9739	0.9538	0.9622	0.9946	0.8836	0.9974	0.9136	0.9800	0.9953	0.9170
	AUC _{TDBS} ↑	0.5055	0.4212	0.3103	0.3736	0.5936	0.3673	0.5757	0.4361	0.3725	0.6449
	AUC _{ODP} ↑	1.5053	1.4076	1.2873	1.3720	1.5759	1.3664	1.5731	1.4352	1.3710	1.6447
Pavia	AUC _(D,F) ↑	0.9887	0.9844	0.9926	0.9879	0.9914	0.9863	0.9925	0.9909	0.9922	0.9970
	AUC _(D,r) ↑	0.1533	0.1003	0.2110	0.1026	0.2145	0.0933	0.5134	0.1719	0.1308	0.1371
	AUC _(F,r) ↓	0.0259	0.0016	0.0048	0.0017	0.0047	0.0019	0.0720	0.0067	0.0004	0.0018
	AUC _{TD} ↑	1.1419	1.0847	1.2036	1.0904	1.2059	1.0795	1.5059	1.1628	1.1230	1.1341
	AUC _{BS} ↑	0.9628	0.9828	0.9878	0.9861	0.9867	0.9844	0.9206	0.9842	0.9917	0.9952
	AUC _{TDBS} ↑	0.1274	0.0987	0.2061	0.1009	0.2098	0.0914	0.4414	0.1652	0.1303	0.1353
	AUC _{ODP} ↑	1.1161	1.0831	1.1987	1.0887	1.2012	1.0777	1.4339	1.1561	1.1225	1.1323

Table 4

Running time of different HAD methods on five datasets.

Time (s)	Method GRX	LEBSR	GAED	Auto-AD	RGAE	BockNet	DFAN	BSDM	NL2Net	HFGDM
Texas Coast I	0.7813	1.6735	27.8371	75.0214	84.1985	1950.1434	37.4242	159.1114	1942.6310	411.6227
Texas Coast II	0.7764	1.3386	24.5288	65.2275	84.1875	1208.5354	37.5630	157.8027	1936.0849	414.7468
Los Angeles	0.7166	1.1395	26.3242	67.1212	85.3791	1315.9141	36.4176	158.5998	2020.8967	425.2413
Bay Champagne	0.6792	1.6396	27.1484	78.9328	78.2781	1185.9172	35.9849	157.5184	1895.6205	373.1541
Pavia	0.8174	2.8042	19.5643	86.0697	43.1057	1085.2153	28.2155	136.5051	1857.1636	193.3957

anomaly detection methods is shown in Table 4. As shown in Table 4, the GRX method achieved the minimum running time on five datasets. The running time of the proposed HFGDM method on five datasets is in the middle of the running time of all compared methods.

The running time of the proposed HFGDM method is relatively high. The reasons are as follows. First, in the forward diffusion process, the low-frequency guided features were extracted from both the input HSI and the corresponding noise HSI at each time step using the LFGM. This processing was performed on each band of these HSI, which incurred notable computational overhead. Second, during training, the proposed diffusion model employed 1000 time steps, resulting in some computational demands. Third, we adopted an AE with fully connected layers as the backbone network. This architecture demonstrated higher computational costs compared to a convolutional neural network (CNN) with the same architecture. Finally, we only used the CPU for the experiment. In the future, optimization strategies we may adopt include reducing the dimensionality of the input HSI, using a lightweight CNN, adopting fewer time steps, and GPU acceleration.

4.7. Ablation study

To verify the effectiveness of the proposed framework, we conducted ablation experiments on the two modules proposed, including the LFGM and the HAFGM. The AUC_(D,F) value is used as an evaluation metric. The impact of different module combinations on detection performance is

Table 5

Detection performance comparison under different combinations of the LFGM and the HAFGM in the proposed method.

LFGM	HAFGM	Dataset Texas Coast I	Texas Coast II	Los Angeles	Bay Champagne	Pavia
×	×	0.9834	0.9925	0.9932	0.9806	0.9823
×	✓	0.9945	0.9948	0.9954	0.9996	0.9959
✓	×	0.9904	0.9939	0.9944	0.9971	0.9927
✓	✓	0.9957	0.9995	0.9960	0.9998	0.9970

shown in Table 5. According to Table 5, when the proposed method includes both the LFGM and the HAFGM, the proposed method on all five datasets achieved the best detection performance. The performance degradation when removing the LFGM and the HAFGM proves that the priori of low-frequency structure and high-frequency details and anomaly features are crucial. In the absence of the LFGM, the model learns from pure noise samples and lacks clear construction guidance. As a result, the reconstruction effect of HSI structure is poor, and the separability of anomaly background is reduced. Without the HAFGM, the model only relies on the noise prediction of the reverse diffusion process for reconstruction, and lacks explicit guidance for high-frequency edge details and anomaly target features in the HSI. Therefore, in the absence of the HAFGM, the fidelity of the background details of the generated HSI is low, and the anomaly target is not prominent. This experimental result indicates that both the proposed LFGM and HAFGM are effective.

To further analyze the influence of edge feature B_{or} and anomaly feature B_{am} in the HAFGM on the detection results, we conducted ablation experiments on the two features. In the experiment, the $AUC_{(D, F)}$ value is used as an evaluation metric. The impact of two features on the detection results under different combinations is shown in Table 6. As shown in Table 6, the proposed method achieves optimal detection performance when both features are present in the HAFGM. Hence, the experimental results demonstrate that both features contribute to improving detection performance.

4.8. Noise robustness analysis

In the proposed method, the detection performance is affected by different noises in the input HSI. To evaluate the noise robustness of the proposed method, Gaussian noise and salt and pepper noise were added to five HSI datasets in this paper. Then, the detection performance of the proposed method is compared on noisy HSI. In the experiment, we added noise to each band of the HSI. The mean and variance of the added Gaussian noise are 0 and 0.01, respectively. The noise intensity of the added salt and pepper noise is 0.01. $AUC_{(D, F)}$ is used to detect performance evaluation metric. Table 7 shows the detection performance of the proposed method under different noise in the input HSI. As shown in Table 7, when Gaussian noise and salt and pepper noise are added to the input HSI, the detection performance of the proposed method decreases to a certain extent compared to the input HSI without noise. However, the proposed method still achieved good detection performance even under the influence of noise. Therefore, the proposed method has a certain ability to suppress noise.

5. Conclusion

This study proposed a novel high-low-frequency feature-guided diffusion model for HAD. The proposed method introduces the global structural information of the HSI and guides the diffusion models to accurately generate HSI and anomaly targets. The proposed method effectively improves anomaly detection performance. First, we constructed an LFGM in the forward diffusion process of the diffusion models to utilize the global structural information of the HSI while suppressing some noise. Second, we designed an HAFGM during the reverse diffusion process to extract high-frequency edge and anomaly features from the HSI. The extracted features serve as explicit priors that guide the network in preserving structural consistency while amplifying rare anomaly patterns during generation. Finally, different from the method of extracting anomaly detection result through reconstruction error, we used the dot product of the spectral sum of the generated HSI and the input HSI as the anomaly detection result. The experimental results show that compared with other state-of-the-art methods, the proposed method on the five datasets achieved satisfactory detection results in both qualitative and quantitative metrics. The ablation experiment demonstrates the effectiveness and superiority of the

Table 6

Detection performance comparison under different combinations of the edge feature and anomaly feature in the proposed method.

Edge feature	Anomaly feature	Dataset	Texas Coast I	Texas Coast II	Los Angeles	Bay Champagne	Pavia
B_{or}	B_{am}						
×	×		0.9904	0.9939	0.9944	0.9971	0.9899
×	✓		0.9954	0.9988	0.9958	0.9973	0.9934
✓	×		0.9927	0.9942	0.9946	0.9995	0.9942
✓	✓		0.9957	0.9995	0.9960	0.9998	0.9970

Table 7

Comparison of the detection performance of the proposed method on the original input HSI and the HSI with noise.

Dataset	No noise	Gaussian noise	Salt and pepper noise
Texas Coast I	0.9957	0.9943	0.9929
Texas Coast II	0.9995	0.9979	0.9934
Los Angeles	0.9960	0.9945	0.9942
Bay Champagne	0.9998	0.9981	0.9936
Pavia	0.9970	0.9948	0.9921

proposed framework.

The proposed method has achieved satisfactory detection results, but it has a long running time. In addition, some hyperparameters in the proposed method are set based on experience or detection performance. The detection performance of the proposed method on different datasets is sensitive to the values of parameters. In the future, we will further reduce the running time of the proposed method by reducing the dimensionality of the HSI, adopting GPU acceleration, and adopting a lightweight network structure. We will also investigate a method for adaptive determination of hyperparameters to achieve automatic detection.

CRediT authorship contribution statement

Pei Xiang: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Conceptualization. **Shuxia Qi:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Data curation. **Soon Ki Jung:** Writing – review & editing, Validation, Supervision, Software, Investigation. **Kun Qian:** Validation, Supervision, Data curation. **Kuanhong Cheng:** Visualization, Software, Resources, Formal analysis. **Shitala Prasad:** Validation, Software, Formal analysis, Data curation. **Huixin Zhou:** Supervision, Project administration, Funding acquisition. **Dong Zhao:** Resources, Project administration, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] F. Meng, X. Cheng, H. Wang, Y. Liu, X. Zhang, Z. Xu, Z. Xing, Using a machine learning-based hyperspectral image classification method for stray light pollution level assessment, *Opt. Laser Technol.* 192 (2025) 113935.
- [2] M. Imani, Discriminant analysis-based attention network for hyperspectral target detection, *Opt. Laser Technol.* 182 (2025) 112208.
- [3] W. Rao, Y. Qu, L. Gao, X. Sun, Y. Wu, B. Zhang, Transferable network with Siamese architecture for anomaly detection in hyperspectral images, *Int. J. Appl. Earth Obs. Geoinf.* 106 (2022) 102669.
- [4] D. Zhao, W. Yan, M. You, J. Zhang, P.V. Arun, C. Jiao, Q. Wang, H. Zhou, Hyperspectral anomaly detection based on empirical mode decomposition and local weighted contrast, *IEEE Sens. J.* 24 (20) (2024) 33847–33861.
- [5] X. He, W. An, Y. Wang, Q. Ling, M. Li, Z. Lin, S. Zhou, CWIMamba: Cross-scale windowed integration state space model for hyperspectral anomaly detection, *IEEE Trans. Geosci. Remote Sens.* 63 (2025) 1–20.
- [6] Y. Huo, X. Cheng, S. Lin, M. Zhang, H. Wang, Memory-augmented autoencoder with adaptive reconstruction and sample attribution mining for hyperspectral anomaly detection, *IEEE Trans. Geosci. Remote Sens.* 62 (2024) 1–18.
- [7] H. Su, Z. Wu, H. Zhang, Q. Du, Hyperspectral anomaly detection: a survey, *IEEE Geosci. Remote Sens. Mag.* 10 (1) (2022) 64–90.
- [8] Y. Chen, Q. Yuan, Y. Tang, Y. Xiao, J. He, T. Han, Z. Liu, L. Zhang, SSTTrack: a unified hyperspectral video tracking framework via modeling spectral-spatial-temporal conditions, *Inform. Fusion* 114 (2025) 102658.
- [9] S. Liu, M. Song, Two-orientations finite markov real-time local anomaly detection via pixel-by-pixel processing for hyperspectral imagery, *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.* 17 (2024) 14219–14236.
- [10] L. Ren, D. Wang, L. Gao, M. Wang, M. Huang, Nonlocal and deep priors for hyperspectral anomaly detection, *IEEE Trans. Geosci. Remote Sens.* 63 (2025) 1–15.
- [11] W. Qin, H. Wang, H. Shu, F. Zhang, J. Wang, X. Cao, X. Zhao, G. Vivone, Hyperspectral anomaly detection fused unified nonconvex tensor ring factors regularization, *IEEE Trans. Geosci. Remote Sens.* 63 (2025) 1–21.
- [12] X. He, J. Wu, Q. Ling, Z. Li, Z. Lin, S. Zhou, Anomaly detection for hyperspectral imagery via tensor low-rank approximation with multiple subspace learning, *IEEE Trans. Geosci. Remote Sens.* 61 (2023) 1–17.
- [13] X. Li, F. Xu, J. Zhang, H. Zhang, X. Lyu, F. Liu, H. Gao, A. Kaup, Frequency-guided denoising network for semantic segmentation of remote sensing images, *IEEE Trans. Geosci. Remote Sens.* 64 (2026) 1–17.
- [14] Y. Chen, Q. Yuan, Y. Tang, Y. Xiao, J. He, Z. Liu, SENSE: Hyperspectral video object tracker via fusing material and motion cues, *Inform. Fusion* 109 (2024) 102395.
- [15] Y. Zhang, X. Li, H. Shi, G. Cao, H. Zhao, Conditional variational autoencoder ensemble for noisy hyperspectral image classification, *Opt. Laser Technol.* 195 (2026) 114602.
- [16] Z. Ma, Y. Zhang, Y. Lian, X. Jiang, X. Liu, Z. Cai, Multiscale memory autoencoder and spatial filtering for hyperspectral anomaly detection, *IEEE Geosci. Remote Sens. Lett.* 22 (2025) 1–5.
- [17] J. Zhong, W. Xie, Y. Li, J. Lei, Q. Du, Characterization of background-anomaly separability with generative adversarial network for hyperspectral anomaly detection, *IEEE Trans. Geosci. Remote Sens.* 59 (7) (2021) 6017–6028.
- [18] J. Ho, A. Jain, P. Abbeel, Denoising diffusion probabilistic models, in: *Proc. Conf. Neural Inform. Process. Syst.* 2020 6840–6851.
- [19] J. Ma, W. Xie, Y. Shi, X. Xiang, Y. Li, L. Fang, BSDM: Background suppression diffusion model for hyperspectral anomaly detection, *IEEE Trans. Circuits Syst. Video Technol.* 36 (1) (2026) 190–204.
- [20] W. Chen, X. Zhi, S. Jiang, Y. Huang, Q. Han, W. Zhang, DWSDiff: Dual-window spectral diffusion for hyperspectral anomaly detection, *IEEE Trans. Geosci. Remote Sens.* 63 (2025) 1–17.
- [21] S.I. Reed, X. Yu, Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution, *IEEE Trans. Acoust., Speech, Signal Process.* 38 (1) (1990) 1760–1770.
- [22] S. Lin, M. Zhang, X. Cheng, L. Shi, P. Gamba, H. Wang, Dynamic low-rank and sparse priors constrained deep autoencoders for hyperspectral anomaly detection, *IEEE Trans. Instrum. Meas.* 73 (2024) 1–18.
- [23] Q. Guo, B. Zhang, Q. Ran, L. Gao, J. Li, A. Plaza, Weighted-RXD and linear filter-based RXD: improving background statistics estimation for anomaly detection in hyperspectral imagery, *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.* 7 (6) (2014) 2351–2366.
- [24] Y. Huo, Y. Dong, C. Wang, M. Zhang, H. Wang, Multi-scale memory network with separation training for hyperspectral anomaly detection, *Inform. Process. Manag.* 63 (2026) 104494.
- [25] H. Kwon, N.M. Nasrabadi, Kernel RX-algorithm: a nonlinear anomaly detector for hyperspectral imagery, *IEEE Trans. Geosci. Remote Sens.* 43 (2) (2005) 388–397.
- [26] J. Liu, Z. Hou, W. Li, R. Tao, D. Orlando, H. Li, Multipixel anomaly detection with unknown patterns for hyperspectral imagery, *IEEE Trans. Neural Netw. Learn. Syst.* 33 (10) (2022) 5557–5567.
- [27] T. Guo, Y. Yang, L. He, C. Fu, F. Luo, Anomaly detection of hyperspectral image by coarse-to-fine tensor two-level decomposition, *IEEE Geosci. Remote Sens. Lett.* 22 (2025) 1–5.
- [28] D. Zhao, X. Xu, M. You, P.V. Arun, Z. Zhao, J. Ren, L. Wu, H. Zhou, Local sub-block contrast and spatial-spectral gradient features fusion for hyperspectral anomaly detection, *Remote Sens.* 17 (4) (2025) 695.
- [29] L. Ren, Z. Han, L. Gao, T. Zhang, R. Wu, H. Zhang, Advances in hyperspectral image unmixing: from algorithmic frameworks to practical applications, *Inform. Geogr.* 2 (1) (2026) 100035.
- [30] Y. Chen, N.M. Nasrabadi, T.D. Tran, Sparse representation for target detection in hyperspectral imagery, *IEEE J. Sel. Topics Signal Process.* 5 (3) (2011) 629–640.
- [31] J. Li, H. Zhang, L. Zhang, L. Ma, Hyperspectral anomaly detection by the use of background joint sparse representation, *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.* 8 (6) (2015) 2523–2533.
- [32] W. Li, Q. Du, Collaborative representation for hyperspectral anomaly detection, *IEEE Trans. Geosci. Remote Sens.* 53 (3) (2015) 1463–1474.
- [33] Y. Huo, S. Wang, C. Wang, M. Zhang, H. Wang, Dual-stream background modeling network with anomaly suppression for hyperspectral anomaly detection, *Int. J. Appl. Earth Obs. Geoinf.* 148 (2026) 105233.
- [34] S. Chang, P. Ghamisi, Nonnegative-constrained joint collaborative representation with union dictionary for hyperspectral anomaly detection, *IEEE Trans. Geosci. Remote Sens.* 60 (2022) 1–13.
- [35] W. Qin, H. Wang, F. Zhang, W. Ma, J. Wang, T. Huang, Nonconvex robust high-order tensor completion using randomized low-rank approximation, *IEEE Trans. Image Process.* 33 (2024) 2835–2850.
- [36] Y. Zhang, B. Du, L. Zhang, S. Wang, A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection, *IEEE Trans. Geosci. Remote Sens.* 54 (3) (2016) 1376–1389.
- [37] L. Li, W. Li, Q. Du, R. Tao, Low-rank and sparse decomposition with mixture of Gaussian for hyperspectral anomaly detection, *IEEE Trans. Cybern.* 51 (9) (2021) 4363–4372.
- [38] Y. Xu, Z. Wu, J. Li, A. Plaza, Z. Wei, Anomaly detection in hyperspectral images based on low-rank and sparse representation, *IEEE Trans. Geosci. Remote Sens.* 54 (4) (2016) 1990–2000.
- [39] L. Li, Z. Wu, B. Wang, Hyperspectral anomaly detection via merging total variation into low-rank representation, *IEEE J. Sel. Topics Signal Process.* 17 (2024) 14894–14907.
- [40] T. Guo, L. He, F. Luo, X. Gong, L. Zhang, X. Gao, Learnable background endmember with subspace representation for hyperspectral anomaly detection, *IEEE Trans. Geosci. Remote Sens.* 62 (2024) 1–13.
- [41] M. Wang, Q. Wang, D. Hong, S.K. Roy, J. Chanussot, Learning tensor low-rank representation for hyperspectral anomaly detection, *IEEE Trans. Cybern.* 53 (1) (2023) 679–691.
- [42] L. Li, W. Li, Y. Qu, C. Zhao, R. Tao, Q. Du, Prior-based tensor approximation for anomaly detection in hyperspectral imagery, *IEEE Trans. Neural Netw. Learn. Syst.* 33 (3) (2022) 1037–1050.
- [43] W. Kong, D. Shen, M. Li, X. Ma, H. Wang, Hyperspectral anomaly detection based on tensor approximation with tensor double nuclear norm, *IEEE Trans. Geosci. Remote Sens.* 63 (2025) 1–14.
- [44] S. Chang, B. Du, L. Zhang, A sparse autoencoder based hyperspectral anomaly detection algorithm using residual of reconstruction error, in: *Proc. IEEE Int. Geosci. Remote Sens. Symp.* (2019) 5488–5491.
- [45] L. Ren, L. Gao, M. Wang, X. Sun, J. Chanussot, HADGSM: a unified nonconvex framework for hyperspectral anomaly detection, *IEEE Trans. Geosci. Remote Sens.* 62 (2024) 1–15.
- [46] W. Qin, H. Wang, F. Zhang, J. Wang, X. Cao, X. Zhao, Tensor ring decomposition-based generalized and efficient nonconvex approach for hyperspectral anomaly detection, *IEEE Trans. Geosci. Remote Sens.* 62 (2024) 1–18.
- [47] X. Cao, Y. Lian, J. Li, K. Wang, C. Ma, Unsupervised multi-level spatio-spectral fusion transformer for hyperspectral image super-resolution, *Opt. Laser Technol.* 176 (2024) 111032.
- [48] G. Fan, Y. Ma, X. Mei, F. Fan, J. Huang, J. Ma, Hyperspectral anomaly detection with robust graph autoencoders, *IEEE Trans. Geosci. Remote Sens.* 60 (2022) 1–14.
- [49] S. Wang, X. Wang, L. Zhang, Y. Zhong, Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder, *IEEE Trans. Geosci. Remote Sens.* 60 (2022) 1–14.
- [50] P. Xiang, S. Ali, S.K. Jung, H. Zhou, Hyperspectral anomaly detection with guided autoencoder, *IEEE Trans. Geosci. Remote Sens.* 60 (2022) 1–18.
- [51] N. Huyan, X. Zhang, D. Quan, J. Chanussot, L. Jiao, Cluster-memory augmented deep autoencoder via optimal transportation for hyperspectral anomaly detection, *IEEE Trans. Geosci. Remote Sens.* 60 (2022) 1–16.
- [52] X. Cheng, Y. Huo, S. Lin, Y. Dong, S. Zhao, M. Zhang, H. Wang, Deep feature aggregation network for hyperspectral anomaly detection, *IEEE Trans. Instrum. Meas.* 73 (2024) 1–16.
- [53] D. Wang, L. Zhuang, L. Gao, X. Sun, M. Huang, A. Plaza, BockNet: Blind-block reconstruction network with a guard window for hyperspectral anomaly detection, *IEEE Trans. Geosci. Remote Sens.* 61 (2023) 1–16.
- [54] D. Wang, L. Ren, X. Sun, L. Gao, J. Chanussot, Nonlocal and local feature-coupled self-supervised network for hyperspectral anomaly detection, *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.* 18 (2025) 6981–6993.
- [55] Z. Wu, B. Wang, Transformer-based autoencoder framework for nonlinear hyperspectral anomaly detection, *IEEE Trans. Geosci. Remote Sens.* 62 (2024) 1–15.
- [56] H. Liu, X. Su, X. Shen, X. Zhou, MSNet: Self-supervised multiscale network with enhanced separation training for hyperspectral anomaly detection, *IEEE Trans. Geosci. Remote Sens.* 62 (2024) 1–13.
- [57] Z. Wang, X. Wang, K. Tan, B. Han, J. Ding, Z. Liu, Hyperspectral anomaly detection based on variational background inference and generative adversarial network, *Pattern Recogn.* 143 (2023) 109795.
- [58] P. Shanmugam, S.A.M.J. Amali, Dual-discriminator conditional generative adversarial network optimized with hybrid manta ray foraging optimization and volcano eruption algorithm for hyperspectral anomaly detection, *Expert Syst. Appl.* 238 (2024) 122058.

- [59] K. Jiang, W. Xie, Y. Li, J. Lei, G. He, Q. Du, Semisupervised spectral learning with generative adversarial network for hyperspectral anomaly detection, *IEEE Trans. Geosci. Remote Sens.* 58 (7) (2020) 5224–5236.
- [60] J. Wang, S. Guo, Z. Hua, R. Huang, J. Hu, M. Gong, CL-CaGAN: Capsule differential adversarial continual learning for cross-domain hyperspectral anomaly detection, *IEEE Trans. Geosci. Remote Sens.* 62 (2024) 1–15.
- [61] C. Li, X. Fan, Y. Zhang, R. Liu, H. Gao, S. Xu, Enhanced feature fusion of graph convolutional network and collaborative attention CNN network for hyperspectral image classification, *Opt. Laser Technol.* 197 (2026) 114601.
- [62] X. Fu, S. Jia, L. Zhuang, M. Xu, J. Zhou, Q. Li, Hyperspectral anomaly detection via deep plug-and-play denoising CNN regularization, *IEEE Trans. Geosci. Remote Sens.* 59 (11) (2021) 9553–9568.
- [63] L. Ren, D. Wang, L. Gao, M. Wang, M. Huang, H. Zhang, HADDNLP: Hyperspectral anomaly detection via double nonlocal priors, *Pattern Recogn.* 172 (2026) 112535.
- [64] Q. Guo, Y. Cen, L. Zhang, Y. Zhang, S. Hu, X. Liu, Robust multi-stage progressive autoencoder for hyperspectral anomaly detection, *Int. J. Appl. Earth Obs. Geoinf.* 134 (2024) 104200.
- [65] Y. Wu, Y. Meng, L. Sun, Diffusing background dictionary for hyperspectral anomaly detection, in: *Proc. Asian Conf. Comput. Vis.* (2024) 1046–1064.
- [66] Z. Yan, L. Li, W. Li, An improved background suppression diffusion model for hyperspectral anomaly detection, in: *Proc. Int. Conf. Comput., Internet Things Smart City* (2024) 103–107.
- [67] P. Xiang, J. Zhang, S. Qi, S.K. Jung, H. Zhou, D. Zhao, Hyperspectral anomaly detection using Taylor expansion and weighted irregular block filter, *Infrared Phys. Technol.* 150 (2025) 105942.
- [68] F. Qian, L. Ma, K. Su, Z. Zhang, Z. Chen, J. Zhang, WDUIE: wavelet-based diffusion for semi-supervised underwater image enhancement, *Opt. Laser Technol.* 192 (2025) 113988.
- [69] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, J. Chanussot, Model-informed multistage unsupervised network for hyperspectral image super-resolution, *IEEE Trans. Geosci. Remote Sens.* 62 (2024) 1–17.
- [70] C.I. Chang, An effective evaluation tool for hyperspectral target detection: 3D receiver operating characteristic curve analysis, *IEEE Trans. Geosci. Remote Sens.* 59 (6) (2021) 5131–5153.
- [71] T. Guo, Y. Yang, F. Luo, C. Fu, L. Zhang, D3Former: Dual-decoder dual-transformer reconstruction network for hyperspectral anomaly detection, *IEEE Trans. Geosci. Remote Sens.* 63 (2025) 1–15.